
EMCH 574 – Vibration and Waves

Homework 01

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Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement [reference](#).
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) [reference](#).
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the *y-axis* scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 Pendulum

A.1 Analyze Pendulum

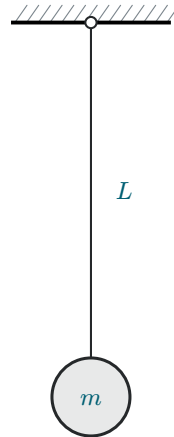


Figure 1: Pendulum of length L and mass m .

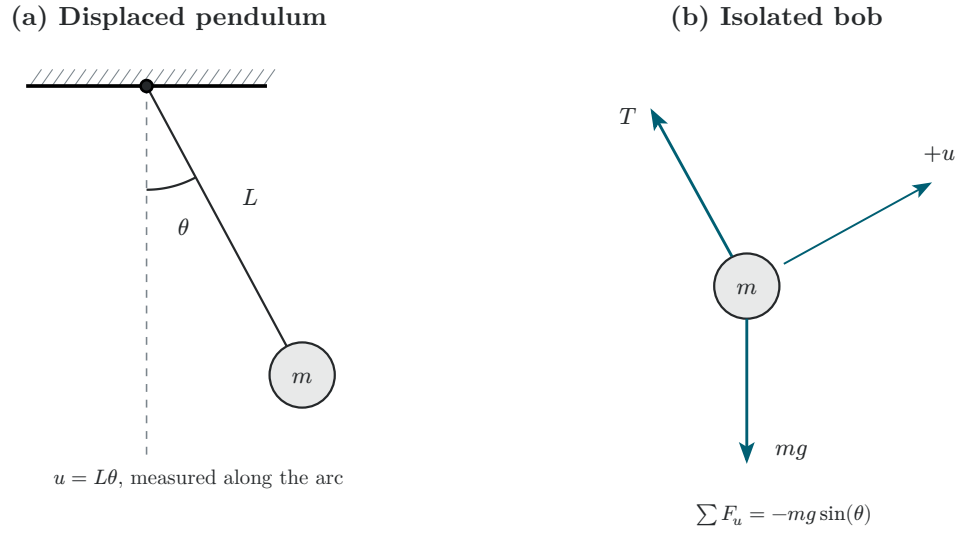
Consider a pendulum of length L and mass m as shown in Figure 1.

Do the following.

- (1) Draw the pendulum displaced by an angle θ from the vertical.
- (2) Draw free-body diagram (FBD).
- (3) Write Newton's law of motion (NLM) equation.
- (4) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (5) Deduce the governing equation.
- (6) Assume $\theta(t) = e^{st}$ and substitute into the governing equation.
- (7) Deduce and solve the characteristic equation.
- (8) What type are the roots $s_{1,2}$? Choose one of the following options:
 - (a) Real.
 - (b) Imaginary.
 - (c) Complex.
- (9) Define the natural frequency ω_n .
- (10) Express the solutions s_1 and s_2 of the characteristic equation in terms of ω_n .
- (11) Write the solution in terms of exponential functions of $\omega_n t$.
- (12) Write the solution in terms of trigonometric functions of $\omega_n t$.
- (13) Write the Hz frequency f_n .

- (14) Write the period of oscillation τ .
- (15) Write the particle motion $u(t)$ in terms of:
- (a) Exponential functions.
 - (b) Trigonometric functions.
- (16) Solve the initial value problem (IVP) and find the response $u(t)$ for:
- (i) Initial displacement u_0 .
 - (ii) Initial velocity $\dot{u}_0$.
 - (iii) Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to A.1

**Figure 2:** Free-body diagram for Problem A.1.

Take positive θ in the direction shown and let $u = L\theta$ be the tangential arc displacement. Then $\ddot{u} = L\ddot{\theta}$. Assume a point mass, a massless inextensible rod or taut string of length L , a fixed frictionless pivot, uniform gravity, planar motion, and negligible air resistance.

The weight is mg , and the tension T has no tangential component. Therefore, the tangential force is the restoring component of gravity,

$$\sum F_u = -mg \sin \theta = m\ddot{u} = mL\ddot{\theta} \quad (\text{A.1.3})$$

For small angular displacement in radians, $\sin \theta \approx \theta$. Substituting this approximation gives the equation of motion

$$mL\ddot{\theta} + mg\theta = 0 \quad (\text{A.1.4})$$

Dividing by mL gives the governing equation

$$\ddot{\theta} + \frac{g}{L}\theta = 0 \quad (\text{A.1.5})$$

To solve the governing equation, assume $\theta = e^{st}$. Then $\ddot{\theta} = s^2 e^{st}$, so substitution gives

$$\left(s^2 + \frac{g}{L}\right) e^{st} = 0.$$

Because $e^{st} \neq 0$, the characteristic equation is

$$\boxed{s^2 + \frac{g}{L} = 0} \quad (\text{A.1.7})$$

Since $g/L > 0$, solving this equation gives purely imaginary roots. Therefore, the answer to item 8 is

$$\boxed{\text{Imaginary}} \quad (\text{A.1.8.b})$$

Define the natural angular frequency by $\omega_n^2 = g/L$. The natural frequency is therefore

$$\boxed{\omega_n = \sqrt{\frac{g}{L}}} \quad (\text{A.1.9})$$

Using this definition, the roots become

$$\boxed{s_1 = i\omega_n, \quad s_2 = -i\omega_n} \quad (\text{A.1.10})$$

The two roots produce the exponential angular response

$$\boxed{\theta(t) = D_1 e^{i\omega_n t} + D_2 e^{-i\omega_n t}} \quad (\text{A.1.11})$$

Using Euler's identity, the same response can be written as

$$\boxed{\theta(t) = A_\theta \cos(\omega_n t) + B_\theta \sin(\omega_n t)} \quad (\text{A.1.12})$$

Because $\omega_n = 2\pi f_n$ and $f_n = 1/\tau$, the frequency and period are

$$\boxed{f_n = \frac{\omega_n}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{g}{L}}} \quad (\text{A.1.13})$$

$$\boxed{\tau = \frac{1}{f_n} = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{L}{g}}} \quad (\text{A.1.14})$$

Multiplying the angular response by L gives the particle-motion response. In exponential form, this is

$$\boxed{u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t}} \quad (\text{A.1.15.a})$$

Euler's identity also gives the trigonometric forms

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) = C \sin(\omega_n t + \varphi) \quad (\text{A.1.15.b})$$

For real A and B , the amplitude and phase are $C = \sqrt{A^2 + B^2}$ and $\varphi = \text{atan2}(A, B)$.

Differentiate the trigonometric response and evaluate it at $t = 0$. The initial conditions give

$$u(0) = A = u_0, \quad \dot{u}(0) = \omega_n B = v_0, \quad v_0 = \dot{u}_0.$$

With initial displacement only, $v_0 = 0$, so

$$u(t) = u_0 \cos(\omega_n t) \quad (\text{A.1.16.i})$$

With initial velocity only, $u_0 = 0$, so

$$u(t) = \frac{v_0}{\omega_n} \sin(\omega_n t) \quad (\text{A.1.16.ii})$$

With both initial conditions applied, the response is

$$u(t) = u_0 \cos(\omega_n t) + \frac{v_0}{\omega_n} \sin(\omega_n t) \quad (\text{A.1.16.iii})$$

The corresponding exponential constants are

$$C_1 = \frac{1}{2} \left(u_0 - \frac{iv_0}{\omega_n} \right), \quad C_2 = \frac{1}{2} \left(u_0 + \frac{iv_0}{\omega_n} \right).$$

The mass cancels when the governing equation is divided by m , so the ideal small-angle pendulum frequency depends on g and L , but not on m .

A.2 Calculate Pendulum Response to Initial Displacement

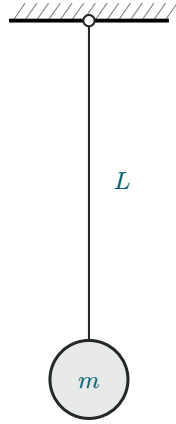


Figure 3: Pendulum of length L and mass m .

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L_{\text{nominal}} = 565 \text{ mm}$ and mass $m = 52 \text{ kg}$ as shown in Figure 3. When manufactured in series production, the pendulum lengths are not exact because of manufacturing tolerances. The lengths of the ten pendulum sets are

$$L = [561.535, 562.691, 563.845, 563.845, 565.000, \\ 565.000, 566.155, 566.155, 567.309, 568.464] \text{ mm}.$$

Do the following.

- (1) For the ten pendulum sets, calculate.
 - (a) Natural frequency ω_n in rad/s.
 - (b) Hz natural frequency f_n .
 - (c) Period of oscillation τ .
 - (d) Display in table format L, ω_n, f_n, τ .
- (2) Mean values: calculate and display the mean values of ω_n, f_n, τ .
- (3) Initial value problem (IVP). Assume initial displacement $u_0 = 2 \text{ mm}$ and zero initial velocity. Use the mean values of ω_n, f_n , and τ to complete items (a)–(c).
 - (a) Calculate and display the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
 - (b) Calculate and display the duration $t_{N_{\text{osc}}}$ of N_{osc} oscillations and the corresponding position of the mass $u_{N_{\text{osc}}}$ for $N_{\text{osc}} = 10$.
 - (c) Plot the response of the system for N_{osc} oscillations and use a data tip to find $u(t_1)$ at t_1 .

Solution to A.2

Use SI units by converting each length from millimetres to metres. For set j , the pendulum relation from A.1 gives

$$\omega_{n,j} = \sqrt{\frac{g}{L_j}} \quad (\text{A.2.1.a})$$

The frequency and period follow from $\omega_{n,j} = 2\pi f_{n,j}$ and $f_{n,j} = 1/\tau_j$, so

$$f_{n,j} = \frac{\omega_{n,j}}{2\pi} \quad (\text{A.2.1.b})$$

$$\tau_j = \frac{2\pi}{\omega_{n,j}} \quad (\text{A.2.1.c})$$

Using $g = 9.81 \text{ m/s}^2$ and the ten lengths gives the following values for A.2.1.d.

Set	L (mm)	ω_n (rad/s)	f_n (Hz)	τ (s)
1	561.535	4.18	0.665	1.503
2	562.691	4.18	0.665	1.505
3	563.845	4.17	0.664	1.506
4	563.845	4.17	0.664	1.506
5	565.000	4.17	0.663	1.508
6	565.000	4.17	0.663	1.508
7	566.155	4.16	0.663	1.509
8	566.155	4.16	0.663	1.509
9	567.309	4.16	0.662	1.511
10	568.464	4.15	0.661	1.513

Compute the mean of each calculated column using $\bar{q} = \frac{1}{10} \sum_{j=1}^{10} q_j$. The mean values required in A.2.2 are

$$\overline{\omega_n} = 4.17 \text{ rad/s} \quad (\text{A.2.2})$$

$$\overline{f_n} = 0.663 \text{ Hz} \quad (\text{A.2.2})$$

$$\overline{\tau} = 1.508 \text{ s} \quad (\text{A.2.2})$$

For A.2.3, the initial velocity is zero, so the A.1 response reduces to

$$u(t) = u_0 \cos(\overline{\omega_n} t) = 2 \cos(4.166893t) \text{ mm} \quad (\text{A.2.3})$$

At $t_1 = 10$ s, substitution into the response gives

$$u(t_1) = 2 \cos(4.166893 \times 10) \text{ mm} = -1.352 \text{ mm} \quad (\text{A.2.3.a})$$

For ten oscillations, multiply the mean period by $N_{\text{osc}} = 10$. The duration and corresponding position are

$$t_{N_{\text{osc}}} = 10\bar{\tau} = 15.08 \text{ s} \quad (\text{A.2.3.b})$$

$$u_{N_{\text{osc}}} = u(10\bar{\tau}) = 2.000 \text{ mm} \quad (\text{A.2.3.b})$$

Plot the response over $0 \leq t \leq 10\bar{\tau}$ using the boxed A.2.3 expression. The figure below shows the ten-oscillation response and marks the value at $t_1 = 10$ s. The plotted value is

$$u(10) = -1.352 \text{ mm} \quad (\text{A.2.3.c})$$

A2. Pendulum released from 2 mm

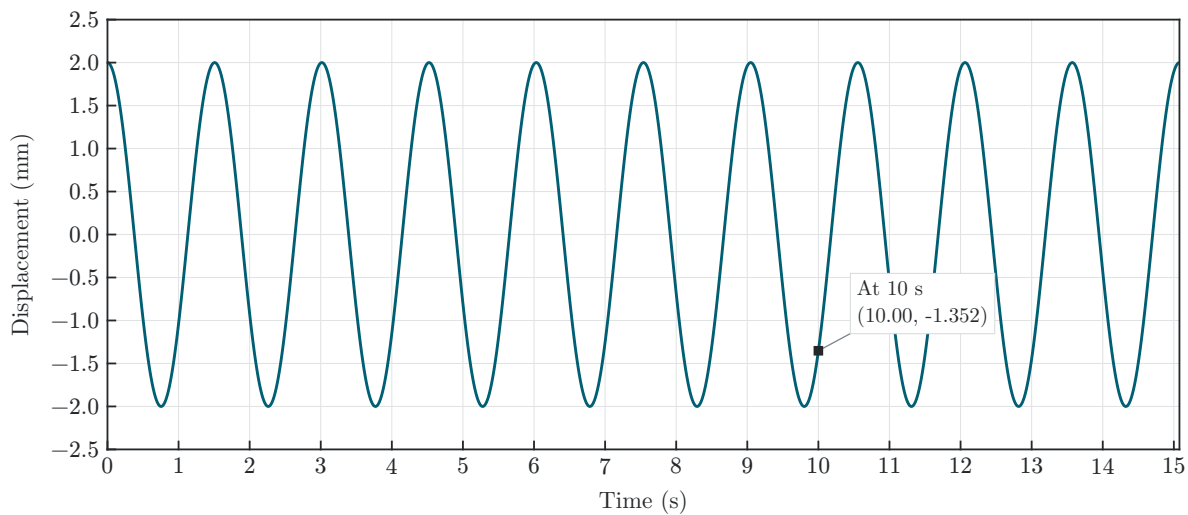


Figure 4: A.2.3.c. Pendulum response over ten oscillations with a data tip at $t = 10$ s.

MATLAB code. *Script used to reproduce Figure 4.*

```
01 u0 = 2;  
02 wn = 4.166893;  
03 t1 = 10;  
04 Nosc = 10;  
05  
06 tau = 2*pi/wn;  
07 t = linspace(0,Nosc*tau,1000);  
08 u = u0*cos(wn*t);  
09  
10 plot(t,u,t1,u0*cos(wn*t1),'ks');  
11 xlabel('Time (s)');  
12 ylabel('u (mm)');
```

A.3 Calculate Pendulum Response to Impact

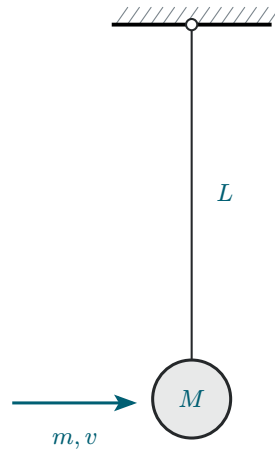


Figure 5: Impact excitation of a pendulum system.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a pendulum of nominal length $L = 565 \text{ mm}$ and mass $M = 52 \text{ kg}$ as shown in Figure 5. The pendulum is initially at rest hanging vertically. A small projectile of mass $m = 22 \text{ g}$ and velocity $v = 945 \text{ m/s}$ hits the mass M and remains embedded into it. This impact imparts an initial velocity to the system that starts to oscillate.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/sec natural frequency ω_n and period of oscillation τ of the pendulum with projectile embedded into it.
- (3) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (4) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (5) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (4).
- (6) Discuss your results.

Solution to A.3

The input values required for A.3.1 are shown below. Convert them to SI units before calculating the response. The projectile mass is denoted by m_p so that it is distinct from the pendulum mass M .

Quantity	Value	Unit
g	9.81	m/s ²
L	0.565	m
M	52.0	kg
m_p	0.0220	kg
v_p	945	m/s
t_1	10.0	s
N_{osc}	10	1

For the pendulum, the small-angle restoring motion gives the undamped equation

$$\ddot{u} + \frac{g}{L}u = 0.$$

Therefore, the natural frequency does not depend on the total mass, because the mass cancels from the ideal pendulum equation. The natural frequency and period are

$$\omega_n = \sqrt{\frac{g}{L}} = 4.16687 \text{ rad/s} \quad (\text{A.3.2})$$

$$\tau = \frac{2\pi}{\omega_n} = 1.50789 \text{ s} \quad (\text{A.3.2})$$

During the short impact, the projectile embeds in the pendulum mass, so linear momentum is conserved. Since the pendulum mass starts at rest, the common velocity immediately after impact satisfies

$$m_p v_p + M(0) = (M + m_p) \dot{u}_0.$$

Solving for the initial velocity gives

$$\dot{u}_0 = \frac{m_p v_p}{M + m_p} = 0.399639 \text{ m/s} \quad (\text{A.3.3})$$

The pendulum is still vertical immediately after impact, so $u_0 = 0$. Substitution into the undamped initial-value solution gives

$$u(t) = u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t).$$

Because $u_0 = 0$, the response becomes

$$u(t) = \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) = 95.9085 \sin(4.16687t) \text{ mm} \quad (\text{A.3.4})$$

At $t_1 = 10 \text{ s}$, substitution into the response gives

$$u(t_1) = 95.9085 \sin(4.16687 \times 10) \text{ mm} = -70.6461 \text{ mm} \quad (\text{A.3.4})$$

Ten oscillations span the interval

$$t_{\text{end}} = N_{\text{osc}} \tau = 10(1.50789) = 15.0789 \text{ s} \quad (\text{A.3.5})$$

The response reaches its first positive maximum one quarter period after impact. The largest absolute response is therefore

$$|u|_{\text{max}} = \frac{\dot{u}_0}{\omega_n} = 95.9085 \text{ mm} \quad (\text{A.3.5.a})$$

The ten-oscillation plot includes data tips at the first maximum and at $t_1 = 10 \text{ s}$. The plotted readings are

$$u_{\text{max}} = 95.9 \text{ mm} \quad (\text{A.3.5.a})$$

$$u(10 \text{ s}) = -70.6 \text{ mm} \quad (\text{A.3.5.b})$$

A3. Response after an embedded-projectile impact

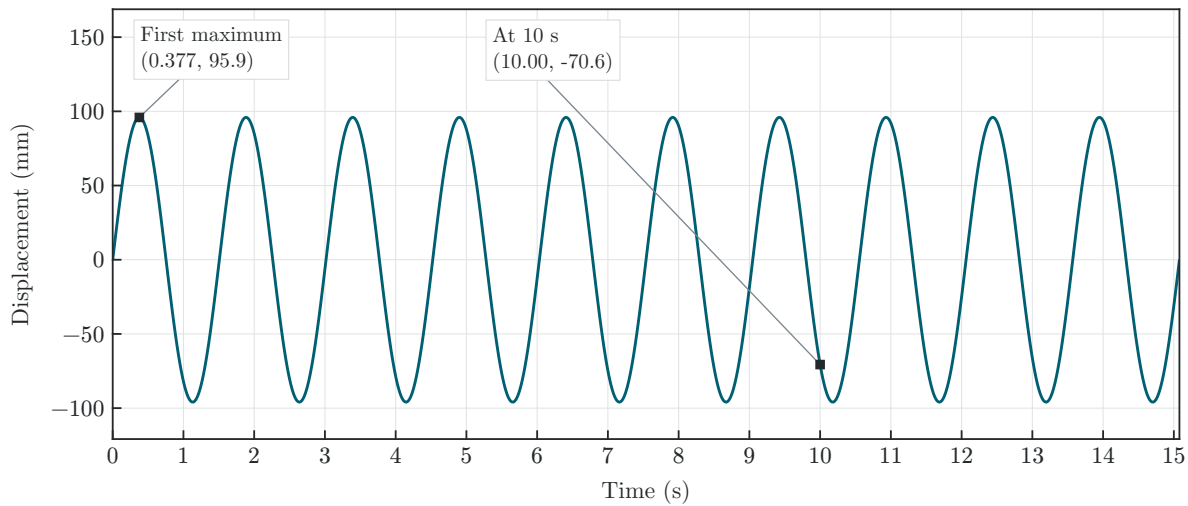


Figure 6: A.3.5.a-b. Impact response with data tips.

The projectile transfers its momentum to the combined mass, which produces a nonzero initial velocity but no initial displacement. The response consequently starts at zero, reaches its first maximum at $t = \tau/4 = 0.377$ s, and then repeats with constant amplitude because damping is neglected. The maximum displacement corresponds to an angular amplitude of approximately $u_{\max}/L = 0.170$ rad = 9.73° , so the small-angle model is reasonable for this calculation. At 10 s, the pendulum is moving through the negative side of the motion rather than being at a maximum, which gives the plotted value of -70.6 mm.

MATLAB code. *Script used to plot the ten-oscillation response.*

```
01 g = 9.81;
02 L = 0.565;
03 M = 52;
04 mp = 0.022;
05 vp = 945;
06 t1 = 10;
07 Nosc = 10;
08 u0 = 0;
09
10 wn = sqrt(g/L);
11 tau = 2*pi/wn;
12 udot0 = mp*vp/(M+mp);
13 A = udot0/wn;
14
15 t = linspace(0,Nosc*tau,1000);
16 u = u0*cos(wn*t) + A*sin(wn*t);
17 u1 = u0*cos(wn*t1) + A*sin(wn*t1);
18
19 plot(t,1000*u,tau/4,1000*A,'ks',t1,1000*u1,'ks');
20 xlabel('Time (s)');
21 ylabel('u (mm)');
22 grid on;
```

2 Spring-Mass-Damper Vibrating Systems

B.1 Analyze Spring-Mass Vibrating System

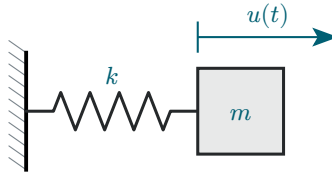


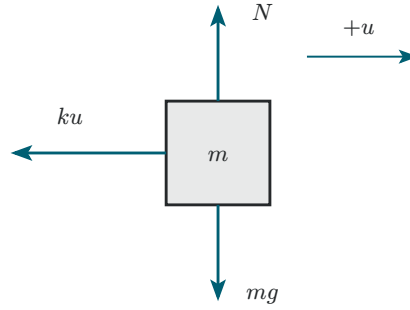
Figure 7: Spring-mass vibrating system.

Consider a spring-mass vibrating system as shown in Figure 7.

Do the following.

- (1) Draw free-body diagram (FBD).
- (2) Write NLM equation.
- (3) Deduce the equation of motion (EOM).
- (4) Deduce the governing equation. Clearly state your assumptions.
- (5) Assume $u(t) = e^{st}$ and substitute into the governing equation.
- (6) Deduce the characteristic equation.
- (7) Define the natural frequency ω_n in terms of m, k .
- (8) Solve the characteristic equation and find $s_{1,2}$ in terms of ω_n .
- (9) What type are the roots $s_{1,2}$? Choose one of the following options:
 - (a) Real.
 - (b) Imaginary.
 - (c) Complex.
- (10) Write the solution in terms of exponential functions.
- (11) Write the solution in terms of trigonometric functions.
- (12) Solve the initial value problem (IVP) and find the response $u(t)$ for:
 - (i) Initial displacement u_0 .
 - (ii) Initial velocity $\dot{u}_0$.
 - (iii) Simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to B.1

**Figure 8:** B.1.1. Free-body diagram for the spring-mass system.

Take displacement $u(t)$ as positive to the right and measure it from the equilibrium position. Assume one degree of freedom, a point mass m , a massless linear spring with stiffness k , frictionless horizontal motion, and no damping or external force. Because the spring force opposes the displacement, its horizontal force is

$$\sum F_x = -ku(t)$$

Applying Newton's second law to the mass gives

$$\boxed{\sum F_x = m\ddot{u}(t)} \quad (\text{B.1.2})$$

Equating the spring force and the inertial force produces the equation of motion,

$$\boxed{m\ddot{u}(t) + ku(t) = 0} \quad (\text{B.1.3})$$

Since m is nonzero, divide by m to obtain the governing equation in normalized form,

$$\boxed{\ddot{u}(t) + \frac{k}{m}u(t) = 0} \quad (\text{B.1.4})$$

Assume an exponential response $u(t) = e^{st}$. Its derivatives are $\dot{u}(t) = se^{st}$ and $\ddot{u}(t) = s^2e^{st}$. Substitution into the equation of motion gives

$$\boxed{(ms^2 + k)e^{st} = 0} \quad (\text{B.1.5})$$

Because e^{st} is never zero, the factor multiplying it must vanish. Therefore, the characteristic equation is

$$ms^2 + k = 0 \quad (\text{B.1.6})$$

Define the undamped natural frequency by

$$\omega_n = \sqrt{\frac{k}{m}} \quad (\text{B.1.7})$$

The characteristic equation then becomes $s^2 + \omega_n^2 = 0$, so its two roots are

$$s_{1,2} = \pm i\omega_n \quad (\text{B.1.8})$$

For positive m and k , these roots have zero real part and nonzero imaginary parts. Thus, the roots are

$$\text{Imaginary} \quad (\text{B.1.9})$$

Using the two distinct roots in the exponential solution gives

$$u(t) = C_1 e^{i\omega_n t} + C_2 e^{-i\omega_n t} \quad (\text{B.1.10})$$

Applying Euler's identity converts this response into the equivalent real trigonometric form,

$$u(t) = A \cos(\omega_n t) + B \sin(\omega_n t) \quad (\text{B.1.11})$$

Let $u_0 = u(0)$ and $\dot{u}_0 = \dot{u}(0)$. Evaluating the trigonometric solution at $t = 0$ gives $A = u_0$. Differentiating and evaluating at $t = 0$ gives $\dot{u}_0 = \omega_n B$, so $B = \dot{u}_0/\omega_n$. Therefore, the three requested initial-condition cases are

$$u(t) = u_0 \cos(\omega_n t) \quad (\text{B.1.12.i})$$

$$u(t) = \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) \quad (\text{B.1.12.ii})$$

$$u(t) = u_0 \cos(\omega_n t) + \frac{\dot{u}_0}{\omega_n} \sin(\omega_n t) \quad (\text{B.1.12.iii})$$

B.2 Analyze Spring-Mass-Damper Vibrating System

Consider the 1-dof damped system consisting of a spring, k , mass, m , and viscous damper, c , as presented in Figure 9. Assume the mass m is displaced from the equilibrium position by a time-dependent displacement $u(t)$.

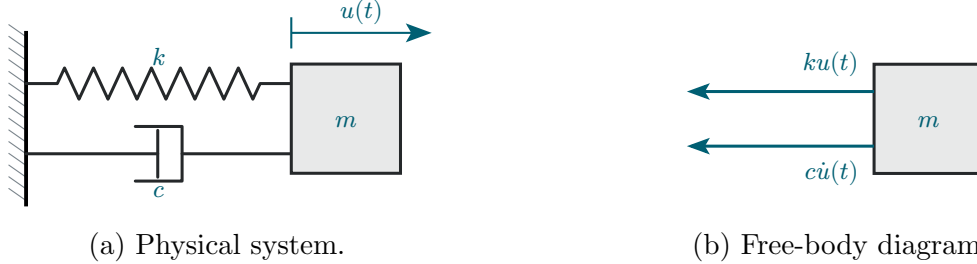


Figure 9: Mass-spring-damper vibrating system.

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k .
- (5) Assume $u(t) = e^{st}$ and substitute into EOM.
- (6) Deduce the characteristic equation.
- (7) Find the roots $s_{1,2}$ of the characteristic equation.
- (8) Define critical damping c_{cr} in terms of m and k .
- (9) Define damping ratio ζ .
- (10) Discuss the values of ζ for the three damping cases:
 - (a) Underdamped.
 - (b) Critically damped.
 - (c) Overdamped.
- (11) Express the physical damping c in terms of damping ratio ζ , mass m , and stiffness k .
- (12) Define the natural frequency ω_n in terms of m and k .
- (13) Express in terms of ω_n , ζ the $s_{1,2}$ roots found at (7).
- (14) Assume underdamped system and express the $s_{1,2}$ roots of (13) as complex numbers in terms of ω_n and ζ .
- (15) Define the damped frequency ω_d .
- (16) Substitute (15) into (14) and express the roots s_1 and s_2 in terms of ω_n , ω_d , ζ .
- (17) Assume underdamped conditions and state the inequality that ζ should satisfy.

- (18) Write the general solution of the ODE deduced at (4) in terms of constants C_1 and C_2 multiplying exponential functions of s_1 and s_2 .
- (19) Use (16) to write the solution (18) in terms of exponential functions of ω_n , ω_d , ζ .
- (20) Rewrite the solution (19) in terms of an exponential decay function multiplying the sum of two complex exponentials.
- (21) Rewrite the solution (20) in terms of an exponential decay function multiplying a trigonometric function and some constants C and ϕ to be determined from the initial conditions.
- (22) Sketch the functions discussed at (21).
- (23) Solve the initial value problem (IVP) for $\zeta < 1$ and find the response $u(t)$ for initial displacement u_0 with zero initial velocity.
- (24) Solve the initial value problem (IVP) and find the response $u(t)$ for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.

Solution to B.2

Take displacement $u(t)$ as positive to the right and measure it from the equilibrium position. Assume one degree of freedom, a point mass m , a massless linear spring with stiffness k , a massless viscous damper with coefficient c , frictionless horizontal motion, and no external force. The damping is nonnegative, so $c \geq 0$.

The free-body diagram for the damped system is shown below. The spring force opposes displacement, and the damper force opposes velocity.

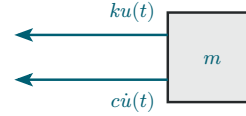


Figure 10: B.2.2. Free-body diagram for the damped system.

Therefore, the spring and damper forces act in the negative x direction, which gives

$$\sum F_x = -c\dot{u}(t) - ku(t) \quad (\text{B.2.2})$$

Applying Newton's law of motion to the mass gives

$$\sum F_x = m\ddot{u}(t) \quad (\text{B.2.3})$$

Equating the force balance to the inertial force and moving all terms to the left gives the equation of motion,

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = 0 \quad (\text{B.2.4})$$

Assume a nontrivial exponential response $u(t) = e^{st}$. Its derivatives are $\dot{u}(t) = se^{st}$ and $\ddot{u}(t) = s^2e^{st}$. Substitution into the equation of motion gives

$$(ms^2 + cs + k)e^{st} = 0 \quad (\text{B.2.5})$$

Because e^{st} is never zero, the remaining factor must vanish. Thus, the characteristic equation is

$$ms^2 + cs + k = 0 \quad (\text{B.2.6})$$

Solving this quadratic equation for s gives the two characteristic roots,

$$s_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m} \quad (\text{B.2.7})$$

Critical damping occurs when the two roots coincide. Therefore, set the discriminant equal to zero,

$$c_{\text{cr}}^2 - 4mk = 0.$$

For nonnegative damping, the critical damping value is

$$\boxed{c_{\text{cr}} = 2\sqrt{mk}} \quad (\text{B.2.8})$$

The damping ratio is the physical damping divided by the critical damping. Hence,

$$\boxed{\zeta = \frac{c}{c_{\text{cr}}} = \frac{c}{2\sqrt{mk}}} \quad (\text{B.2.9})$$

Since $c \geq 0$, the three damping cases are

$$\boxed{0 \leq \zeta < 1} \quad \text{Underdamped} \quad (\text{B.2.10.a})$$

$$\boxed{\zeta = 1} \quad \text{Critically damped} \quad (\text{B.2.10.b})$$

$$\boxed{\zeta > 1} \quad \text{Overdamped} \quad (\text{B.2.10.c})$$

The value $\zeta = 0$ is the undamped limiting case. Using $c = \zeta c_{\text{cr}}$ and the critical damping expression gives the physical damping in terms of ζ , m , and k ,

$$\boxed{c = 2\zeta\sqrt{mk}} \quad (\text{B.2.11})$$

The stiffness-to-mass ratio is the square of the undamped natural frequency. Therefore,

$$\boxed{\omega_n = \sqrt{\frac{k}{m}}} \quad (\text{B.2.12})$$

Using $c/(2m) = \zeta\omega_n$ and $k/m = \omega_n^2$ in the characteristic roots gives

$$\boxed{s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}} \quad (\text{B.2.13})$$

For an underdamped system, $\zeta < 1$, so $\zeta^2 - 1 < 0$. The square root can therefore be written as

$$\sqrt{\zeta^2 - 1} = i\sqrt{1 - \zeta^2}.$$

Substitution into the roots from (B.2.13) gives the complex roots,

$$s_{1,2} = -\zeta\omega_n \pm i\omega_n\sqrt{1-\zeta^2} \quad (\text{B.2.14})$$

Define the damped frequency as the positive imaginary part of the underdamped roots,

$$\omega_d = \omega_n\sqrt{1-\zeta^2} \quad (\text{B.2.15})$$

Substituting this definition into (B.2.14) separates the two roots as

$$\begin{aligned} s_1 &= -\zeta\omega_n + i\omega_d, \\ s_2 &= -\zeta\omega_n - i\omega_d. \end{aligned} \quad (\text{B.2.16})$$

The underdamped condition is therefore

$$\zeta < 1 \quad (\text{B.2.17})$$

With physical nonnegative damping, this is $0 \leq \zeta < 1$. The value $\zeta = 0$ gives the undamped limit, while $0 < \zeta < 1$ gives a decaying oscillation.

Using the two roots in (B.2.16), the general solution is the linear superposition of the two exponential solutions,

$$u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t} \quad (\text{B.2.18})$$

Substituting the underdamped roots gives

$$u(t) = C_1 e^{(-\zeta\omega_n + i\omega_d)t} + C_2 e^{(-\zeta\omega_n - i\omega_d)t} \quad (\text{B.2.19})$$

Because $e^{a+b} = e^a e^b$, factor out the common decay term to obtain

$$u(t) = e^{-\zeta\omega_n t} (C_1 e^{i\omega_d t} + C_2 e^{-i\omega_d t}) \quad (\text{B.2.20})$$

Euler's identities convert the harmonic part into a real trigonometric form. Define $A = C_1 + C_2$ and $B = i(C_1 - C_2)$. Then

$$C_1 e^{i\omega_d t} + C_2 e^{-i\omega_d t} = A \cos(\omega_d t) + B \sin(\omega_d t).$$

Writing this sinusoid in amplitude-phase form gives $A = C \sin \phi$ and $B = C \cos \phi$. Thus,

$$u(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi), \quad C = \sqrt{A^2 + B^2}, \quad \phi = \text{atan2}(A, B) \quad (\text{B.2.21})$$

The sinusoidal carrier is bounded by the two exponential envelopes $u_{\text{env},\pm}(t) = \pm Ce^{-\zeta\omega_n t}$. The carrier keeps the constant damped frequency ω_d , while its amplitude decreases with time. The resulting sketch is shown below using illustrative values $\omega_n = 1$, $\zeta = 0.20$, $C = 1$, and $\phi = 0.5$ radians.

B2. Underdamped response and exponential envelope

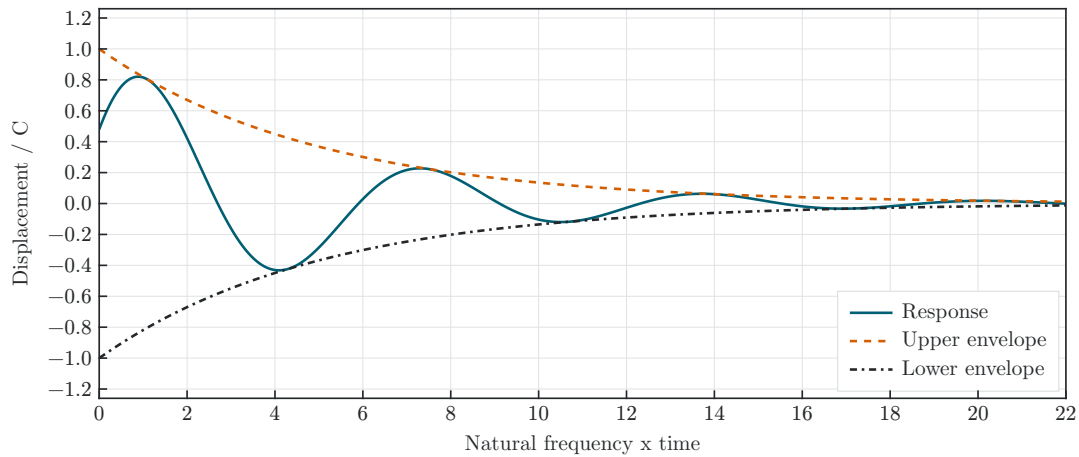


Figure 11: B.2.22. Underdamped oscillation and its exponential decay envelopes.

MATLAB code. Code used to reproduce the B.2.22 response sketch.

```

01 wn = 1; zeta = 0.20; C = 1; phi = 0.5; Nosc = 3;
02 wd = wn*sqrt(1-zeta^2);
03 t = linspace(0,Nosc*2*pi/wd,1000);
04 u = C*exp(-zeta*wn*t).*sin(wd*t+phi);
05 e = C*exp(-zeta*wn*t);
06
07 plot(wn*t,u,'Color',[0 .427 .459]); hold on
08 plot(wn*t,e,'--','Color',[.776 .365 0])
09 plot(wn*t,-e,'-k')
10 xlabel('Natural frequency x time');
11 ylabel('Displacement / C');
12 legend('Response','Upper envelope','Lower envelope');
13 ylim([-1.2 1.2]); grid on

```

For the initial-displacement case, let $u(0) = u_0$ and $\dot{u}(0) = 0$. Evaluating the cosine-sine form at $t = 0$ gives $A = u_0$. Differentiating that form before setting $t = 0$ gives

$$\dot{u}(0) = -\zeta\omega_n A + \omega_d B = 0.$$

Therefore, $B = \zeta\omega_n u_0 / \omega_d$, and the response is

$$u(t) = u_0 e^{-\zeta\omega_n t} \left[\cos(\omega_d t) + \frac{\zeta\omega_n}{\omega_d} \sin(\omega_d t) \right] \quad (\text{B.2.23})$$

For simultaneous initial displacement and velocity, let $u(0) = u_0$ and $\dot{u}(0) = \dot{u}_0$. The same two initial-condition equations give

$$A = u_0, \quad B = \frac{\dot{u}_0 + \zeta\omega_n u_0}{\omega_d}.$$

Substitution into the trigonometric solution gives the complete underdamped IVP response,

$$u(t) = e^{-\zeta\omega_n t} \left[u_0 \cos(\omega_d t) + \frac{\dot{u}_0 + \zeta\omega_n u_0}{\omega_d} \sin(\omega_d t) \right] \quad (\text{B.2.24})$$

B.3 Calculate Spring-Mass-Damper Response to Initial Displacement

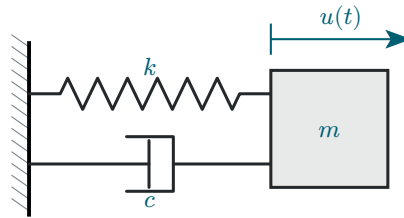


Figure 12: *Spring-mass-damper vibrating system.*

Consider a spring-mass vibrating system with $m = 12 \text{ kg}$, $k = 655 \text{ N/m}$, $c = 19 \text{ N/(m/s)}$ as shown in Figure 12. The vibrating system is given an initial displacement $u_0 = 3.5 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (3) Calculate the critical damping c_{cr} .
- (4) Calculate the damping ratio ζ .
- (5) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (6) Calculate the damped period of oscillation τ_d .
- (7) Calculate the characteristic-equation roots s_1 , s_2 and the constants C_1 , C_2 .
- (8) Calculate the constants A , B and then the constants C , ϕ .
- (9) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (10) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (11) Plot the response of the system for $N_{osc} = 10$ oscillations and use datatips on the plot to find the following:
 - (a) The largest response.
 - (b) The response value calculated at item (9).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (10).
- (12) Discuss your results.

Solution to B.3

The given system has mass $m = 12$ kg, stiffness $k = 655$ N/m, damping coefficient $c = 19$ N s/m, initial displacement $u_0 = 3.5$ mm, and initial velocity $\dot{u}_0 = 0$. The response will be plotted for $N_{\text{osc}} = 10$ damped oscillations.

Quantity	Value	Unit
m	12.0	kg
k	655	N/m
c	19.0	N s/m
u_0	3.50	mm
$\dot{u}_0$	0	mm/s
N_{osc}	10	–

The natural frequency follows from the stiffness-to-mass ratio. Converting it to hertz requires division by 2π .

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{655}{12}} = 7.388 \text{ rad/s}, \quad f_n = \frac{\omega_n}{2\pi} = 1.176 \text{ Hz}.$$

$$\boxed{\omega_n = 7.39 \text{ rad/s}, \quad f_n = 1.176 \text{ Hz}} \quad (\text{B.3.2})$$

Critical damping is the value that makes the characteristic-equation discriminant equal to zero. Therefore,

$$c_{\text{cr}} = 2\sqrt{mk} = 2\sqrt{(12)(655)} = 177.313 \text{ N s/m}.$$

$$\boxed{c_{\text{cr}} = 177.3 \text{ N s/m}} \quad (\text{B.3.3})$$

The damping ratio compares the actual damping with the critical value. Since $c < c_{\text{cr}}$, the system is underdamped.

$$\zeta = \frac{c}{c_{\text{cr}}} = \frac{19}{177.313} = 0.107155.$$

$$\boxed{\zeta = 0.1072 \quad (\text{underdamped})} \quad (\text{B.3.4})$$

For an underdamped system, the oscillatory part of the characteristic roots has frequency $\omega_d = \omega_n \sqrt{1 - \zeta^2}$. Converting this frequency to hertz gives

$$\omega_d = 7.388\sqrt{1 - (0.107155)^2} = 7.346 \text{ rad/s}, \quad f_d = \frac{\omega_d}{2\pi} = 1.169 \text{ Hz}.$$

$$\boxed{\omega_d = 7.35 \text{ rad/s}, \quad f_d = 1.169 \text{ Hz}} \quad (\text{B.3.5})$$

One damped oscillation lasts one damped period. Thus,

$$\tau_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{7.346} = 0.8554 \text{ s}.$$

$$\boxed{\tau_d = 0.855 \text{ s}} \quad (\text{B.3.6})$$

The real part of each root is the exponential decay rate $-\zeta\omega_n = -c/(2m) = -0.791667 \text{ s}^{-1}$. Combining this decay rate with the damped frequency gives

$$s_{1,2} = -\zeta\omega_n \pm i\omega_d = -0.791667 \pm i 7.3455 \text{ s}^{-1}.$$

For the exponential form $u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}$, the real trigonometric coefficients satisfy $A = C_1 + C_2$ and $B = i(C_1 - C_2)$. The initial conditions determine $A = u_0$ and $B = \zeta\omega_n u_0 / \omega_d$.

$$C_1 = \frac{A - iB}{2} = 1.750 - 0.1886i \text{ mm}, \quad C_2 = \frac{A + iB}{2} = 1.750 + 0.1886i \text{ mm}.$$

$$\boxed{\begin{aligned} s_1 &= -0.7917 + i 7.3455 \text{ s}^{-1}, & s_2 &= -0.7917 - i 7.3455 \text{ s}^{-1}, \\ C_1 &= 1.750 - 0.1886i \text{ mm}, & C_2 &= 1.750 + 0.1886i \text{ mm} \end{aligned}} \quad (\text{B.3.7})$$

The zero initial velocity condition fixes the coefficient of the sine term. Substituting the numerical values gives

$$A = u_0 = 3.50 \text{ mm}, \quad B = \frac{\zeta\omega_n u_0}{\omega_d} = 0.3772 \text{ mm}.$$

Writing the harmonic part as a single sine function gives $A = C \sin \phi$ and $B = C \cos \phi$. Therefore,

$$C = \sqrt{A^2 + B^2} = 3.520 \text{ mm}, \quad \phi = \text{atan2}(A, B) = 1.463 \text{ rad}.$$

The displacement response is consequently

$$u(t) = 3.520e^{-0.7917t} \sin(7.3455t + 1.463) \text{ mm.}$$

$$\boxed{\begin{array}{ll} A = 3.50 \text{ mm}, & B = 0.377 \text{ mm}, \\ C = 3.52 \text{ mm}, & \phi = 1.463 \text{ rad} \end{array}} \quad (\text{B.3.8})$$

At one damped period, the sine and cosine terms return to their initial values. The oscillatory factor is therefore one, while the exponential factor has decayed for one period.

$$t_1 = \tau_d = 0.8554 \text{ s}, \quad u_1 = u(t_1) = u_0 e^{-\zeta \omega_n t_1} = 1.778 \text{ mm.}$$

$$\boxed{t_1 = 0.855 \text{ s}, \quad u_1 = 1.778 \text{ mm}} \quad (\text{B.3.9})$$

The displacement magnitude defining the 2% band is

$$u_{0.02} = 0.02u_0 = 0.02(3.5) = 0.0700 \text{ mm.}$$

$$\boxed{u_{0.02} = 0.0700 \text{ mm}} \quad (\text{B.3.10})$$

The largest response occurs at release because the initial displacement is 3.5 mm and the envelope decreases thereafter. The one-period response is the value calculated above. To find the permanent settling time, solve $u(t) = -u_{0.02}$ on the falling side of the last lobe that reaches the lower band. The numerical crossing is $t_s = 4.7867 \text{ s}$. All later extrema have smaller magnitude, so the response remains inside the 2% band after this crossing.

$$\boxed{|u|_{\max} = 3.50 \text{ mm} \quad \text{at } t = 0} \quad (\text{B.3.11.a})$$

$$\boxed{u(t_1) = 1.778 \text{ mm} \quad \text{at } t_1 = 0.855 \text{ s}} \quad (\text{B.3.11.b})$$

$$\boxed{t_s = 4.79 \text{ s}, \quad u(t_s) = -0.0700 \text{ mm}} \quad (\text{B.3.11.c})$$

B3. Displacement release

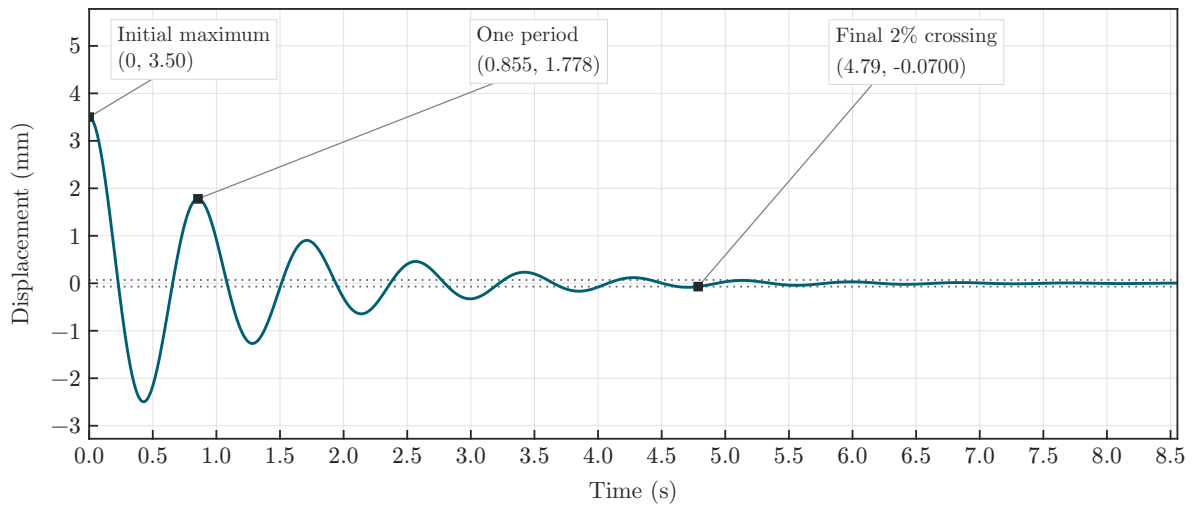


Figure 13: *B.3.11.a-b. Ten-cycle response with the initial maximum, one-period response, and final 2% crossing.*

B3. Final crossing of the 2% band

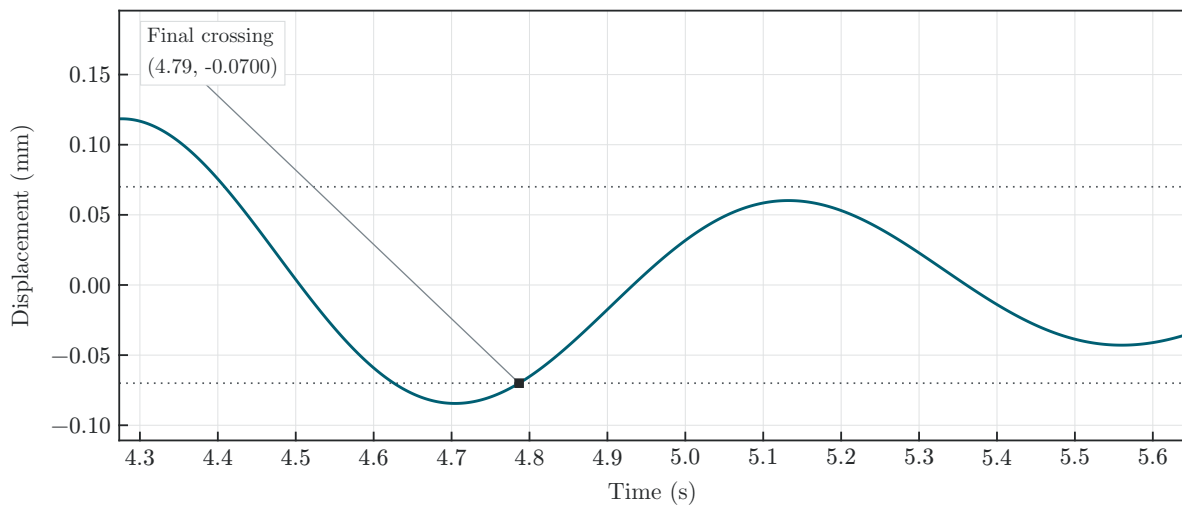


Figure 14: *B.3.11.c. Detail of the final 2% crossing of the settling band.*

MATLAB code. *MATLAB code used to reproduce the B.3.11 response plot.*

```

01 m = 12; k = 655; c = 19; u0 = 3.5; Nosc = 10; ts = 4.786694;
02 wn = sqrt(k/m); zeta = c/(2*sqrt(m*k)); wd = wn*sqrt(1-zeta^2);
03 tau = 2*pi/wd; t = linspace(0,Nosc*tau,2000);
04 u = u0*exp(-zeta*wn*t).*(cos(wd*t)+zeta*wn/wd*sin(wd*t));
05 u02 = .02*u0; u1 = u0*exp(-zeta*wn*tau);
06
07 plot(t,u,'Color',[0 .427 .459]); hold on
08 plot([0 t(end)],[u02 u02],':','Color',[.776 .365 0]);
09 plot([0 t(end)],[u02 -u02],':','Color',[.776 .365 0]);
10 plot([0 tau ts],[u0 u1 -u02],'ks');
11 xlabel('Time (s)'); ylabel('Displacement (mm)');
12 legend('Response','2% band','Location','northeast'); ylim([-3 5]); grid on

```

The plot shows an underdamped response whose oscillation frequency remains constant while its amplitude decreases exponentially. The initial condition produces the largest displacement at release, and the displacement after one period is smaller by the factor $e^{-\zeta\omega_n\tau_d}$. The response first enters the 2% band before 4.79 s, but it does not remain there until the final crossing at $t_s = 4.79$ s. The exponential envelope gives the conservative bound $t_{\text{envelope}} = 4.95$ s, which is later because the response is not at the envelope maximum at the exact crossing.

(B.3.12)

B.4 Calculate Truck Suspension Response

Consider a pickup truck wheel suspension consisting of a leaf spring and an oil damper. The truck and its suspension can be modeled, in first approximation, as a k - m - c spring-mass-damper system as shown in Figure 15.

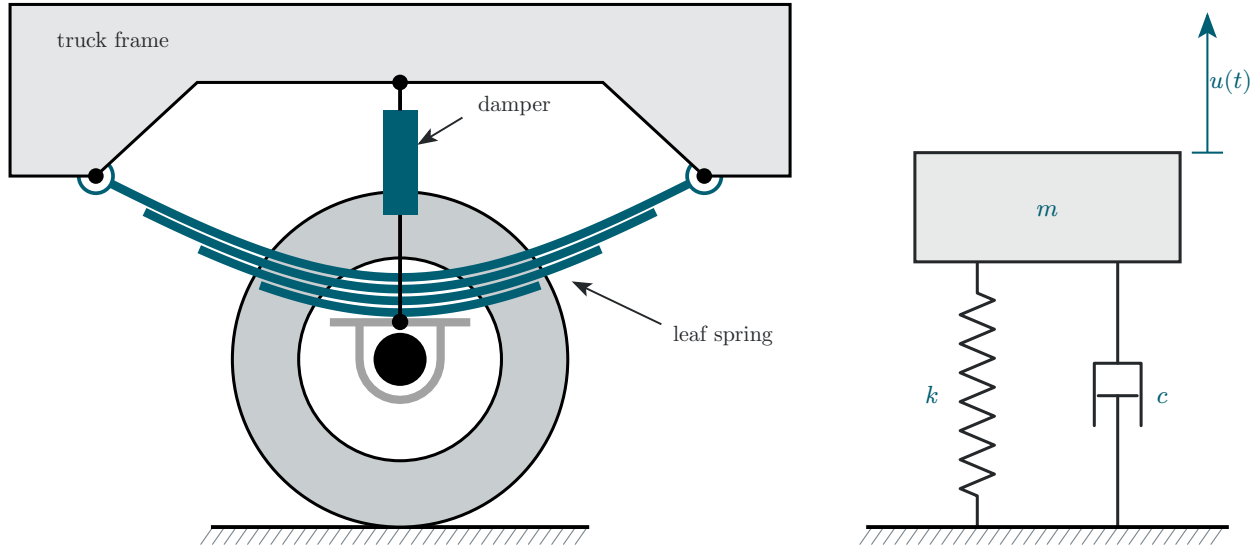


Figure 15: Truck suspension consisting of a leaf spring and an oil damper. The wheel contact is treated as the road datum in the one-degree-of-freedom idealization.

The leaf spring can be approximated by a uniform beam of length $L = 830 \text{ mm}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 65 \text{ mm}$, made of steel with $E = 200 \text{ GPa}$. The truck mass apportioned to the wheel can be estimated as $m = 457 \text{ kg}$. Assume initial displacement $u_0 = 125 \text{ mm}$ and zero initial velocity $\dot{u}_0 = 0$.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I of the beam cross section.
- (3) Calculate the flexural stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n and in Hz f_n .
- (6) Calculate the critical damping c_{cr} .
- (7) Assume a 25% damping ratio, i.e., $\zeta = 0.25$, and calculate the physical damping c . Print ζ and c .
- (8) Calculate the damped frequency in rad/sec ω_d and in Hz f_d .
- (9) Calculate the damped period of oscillation τ_d .
- (10) Calculate the characteristic-equation roots s_1, s_2 and the constants C_1, C_2 .
- (11) Calculate the constants A, B and then the constants C, ϕ .

-
- (12) Calculate the duration t_1 of one oscillation and the corresponding position $u_1 = u(t_1)$.
- (13) Calculate the value $u_{0.02}$ representing 2% of the initial displacement.
- (14) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and use datatips on the plot to find the following:
- (a) The largest response.
 - (b) The response value calculated at item (12).
 - (c) The time when the response magnitude has decreased permanently below $u_{0.02}$ calculated at item (13).
- (15) Discuss your results.
- (16) Graduate work. Assume the damper is worn out such that ζ is reduced to $\zeta = 10\%$.
- (a) Repeat the calculation of items (6) through (12).
 - (b) Compare and discuss the two results.

Solution to B.4

Use SI units for the beam calculation and report displacement in millimeters for the response plots. The leaf spring is idealized as a simply supported beam with a concentrated load at its center. This matches the two spring-eye supports and the center displacement shown in the model. The corresponding center deflection is

$$\delta = \frac{PL^3}{48EI}, \quad k = \frac{P}{\delta} = \frac{48EI}{L^3}.$$

This support condition is important because a cantilever model would produce a different stiffness.

B.4.1. Input data.

Quantity	Value	Unit
L	0.830	m
b	0.065	m
h	0.003	m
E	200	GPa
m	457	kg
u_0	125	mm
$\dot{u}_0$	0	mm/s
ζ_H	0.25	1
ζ_W	0.10	1
g	9.81	m/s ²

For vertical bending about the horizontal centroidal axis, the section dimension perpendicular to that axis is h , so h is cubed in the second moment of area.

$$I = \frac{bh^3}{12} = \frac{(0.065)(0.003)^3}{12} = 1.4625 \times 10^{-10} \text{ m}^4 = 146.25 \text{ mm}^4.$$

$$\boxed{I = 1.4625 \times 10^{-10} \text{ m}^4 = 146.25 \text{ mm}^4} \quad (\text{B.4.2})$$

Multiplying the section property by the elastic modulus gives the flexural rigidity.

$$EI = (200 \times 10^9)(1.4625 \times 10^{-10}) = 29.25 \text{ N m}^2.$$

$$\boxed{EI = 29.25 \text{ N m}^2} \quad (\text{B.4.3})$$

Substitution into the simply supported center-load relation gives the equivalent spring constant.

$$k = \frac{48EI}{L^3} = \frac{48(29.25)}{(0.830)^3} = 2455.46 \text{ N/m}.$$

$$k = 2.455 \times 10^3 \text{ N/m} \quad (\text{B.4.4})$$

The undamped natural frequency follows from the mass and equivalent stiffness. Its cyclic frequency is found by dividing by 2π .

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{2455.46}{457}} = 2.31797 \text{ rad/s}, \quad f_n = \frac{\omega_n}{2\pi} = 0.368917 \text{ Hz}.$$

$$\omega_n = 2.32 \text{ rad/s}, \quad f_n = 0.3689 \text{ Hz} \quad (\text{B.4.5})$$

Critical damping makes the characteristic-equation roots coincide. Therefore,

$$c_{\text{cr}} = 2\sqrt{mk} = 2\sqrt{(457)(2455.46)} = 2118.63 \text{ N s/m}.$$

$$c_{\text{cr}} = 2.119 \times 10^3 \text{ N s/m} \quad (\text{B.4.6})$$

For the original damper, the specified damping ratio is $\zeta_{\text{H}} = 0.25$. The physical damping is the damping ratio multiplied by the critical value.

$$c_{\text{H}} = \zeta_{\text{H}} c_{\text{cr}} = (0.25)(2118.63) = 529.657 \text{ N s/m}.$$

$$\zeta_{\text{H}} = 0.250, \quad c_{\text{H}} = 529.7 \text{ N s/m} \quad (\text{B.4.7})$$

Because $\zeta_{\text{H}} < 1$, the original system is underdamped. Its damped frequency and cyclic frequency are

$$\omega_{d,\text{H}} = \omega_n \sqrt{1 - \zeta_{\text{H}}^2} = 2.31797 \sqrt{1 - 0.25^2} = 2.24437 \text{ rad/s}, \quad f_{d,\text{H}} = \frac{\omega_{d,\text{H}}}{2\pi} = 0.357202 \text{ Hz}.$$

$$\omega_{d,\text{H}} = 2.24 \text{ rad/s}, \quad f_{d,\text{H}} = 0.3572 \text{ Hz} \quad (\text{B.4.8})$$

The time for one damped oscillation is the damped period.

$$\tau_{d,H} = \frac{2\pi}{\omega_{d,H}} = \frac{2\pi}{2.24437} = 2.79954 \text{ s.}$$

$$\boxed{\tau_{d,H} = 2.80 \text{ s}} \quad (\text{B.4.9})$$

Define the decay rate $\alpha_H = \zeta_H \omega_n = 0.579493 \text{ s}^{-1}$. The characteristic roots are then

$$s_{1,2} = -\alpha_H \pm i\omega_{d,H} = -0.579493 \pm i 2.24437 \text{ s}^{-1}.$$

For the zero-velocity initial condition, the exponential constants are obtained from the trigonometric coefficients $A = u_0$ and $B = \alpha_H u_0 / \omega_{d,H}$. Since $A = C_1 + C_2$ and $B = i(C_1 - C_2)$,

$$C_1 = \frac{A - iB}{2} = 62.500 - 16.137i \text{ mm}, \quad C_2 = \frac{A + iB}{2} = 62.500 + 16.137i \text{ mm}.$$

$$\boxed{\begin{aligned} s_1 &= -0.5795 + i 2.2444 \text{ s}^{-1}, & s_2 &= -0.5795 - i 2.2444 \text{ s}^{-1}, \\ C_1 &= 62.50 - 16.14i \text{ mm}, & C_2 &= 62.50 + 16.14i \text{ mm} \end{aligned}} \quad (\text{B.4.10})$$

The real trigonometric coefficients are

$$A = u_0 = 125.0 \text{ mm}, \quad B = \frac{\alpha_H u_0}{\omega_{d,H}} = 32.2749 \text{ mm}.$$

Writing the harmonic part as a single sine function gives $A = C \sin \phi$ and $B = C \cos \phi$. Therefore,

$$C = \sqrt{A^2 + B^2} = 129.099 \text{ mm}, \quad \phi = \text{atan2}(A, B) = 1.31812 \text{ rad}.$$

The displacement response for the original damper is consequently

$$u_H(t) = 129.099 e^{-0.579493t} \sin(2.24437t + 1.31812) \text{ mm}.$$

$$\boxed{\begin{aligned} A &= 125.0 \text{ mm}, & B &= 32.27 \text{ mm}, \\ C &= 129.1 \text{ mm}, & \phi &= 1.318 \text{ rad} \end{aligned}} \quad (\text{B.4.11})$$

At one damped period, the sine and cosine terms return to their initial values. The displacement is therefore the initial displacement multiplied by the decay factor for one period.

$$t_{1,H} = \tau_{d,H} = 2.79954 \text{ s}, \quad u_{1,H} = u_H(t_{1,H}) = u_0 e^{-\alpha_H t_{1,H}} = 24.680 \text{ mm}.$$

$$\boxed{t_{1,H} = 2.80 \text{ s}, \quad u_{1,H} = 24.68 \text{ mm}} \quad (\text{B.4.12})$$

The displacement magnitude defining the 2% band is

$$u_{0.02} = 0.02u_0 = 0.02(125) = 2.50 \text{ mm}.$$

$$\boxed{u_{0.02} = 2.50 \text{ mm}} \quad (\text{B.4.13})$$

The largest response is the initial displacement, so $|u|_{\max} = 125 \text{ mm}$ at $t = 0$. The last extremum with magnitude greater than $u_{0.02}$ occurs at approximately $t = 5.599 \text{ s}$, where $u = 4.873 \text{ mm}$. Solving the descending branch for $u(t) = u_{0.02}$ gives the final entry into the 2% band at $t_s = 6.090 \text{ s}$. All later extrema have magnitude below 2.50 mm .

$$\boxed{|u|_{\max} = 125.0 \text{ mm} \quad \text{at } t = 0} \quad (\text{B.4.14.a})$$

$$\boxed{u(t_{1,H}) = 24.68 \text{ mm} \quad \text{at } t_{1,H} = 2.80 \text{ s}} \quad (\text{B.4.14.b})$$

$$\boxed{t_{s,H} = 6.09 \text{ s}, \quad u_H(t_{s,H}) = +2.50 \text{ mm}} \quad (\text{B.4.14.c})$$

B4 healthy. Displacement release

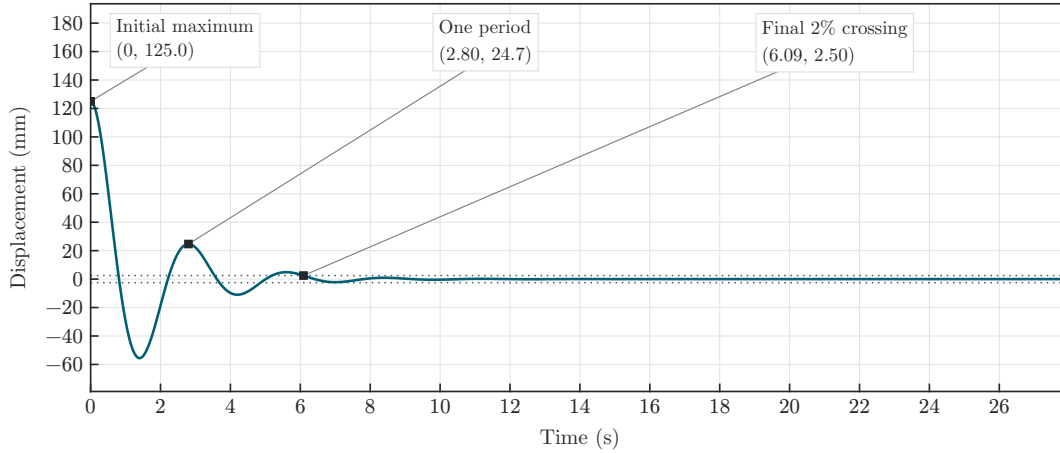


Figure 16: B.4.14.a-b-c. Original damper response for ten oscillations.

B.4.15. Discussion.

The original damper produces a decaying underdamped response. The damped period is 2.80 s, and the displacement after one period is reduced from 125 mm to 24.68 mm. The response remains oscillatory, but its extrema decrease because the envelope is $Ce^{-\alpha_H t}$. The final permanent entry into the 2% band occurs at 6.09 s.

The calculated static displacement under the apportioned weight is

$$\delta_{\text{static}} = \frac{mg}{k} = \frac{(457)(9.81)}{2455.46} = 1.826 \text{ m}.$$

This is larger than the beam span and much larger than the prescribed 125 mm initial displacement. The value indicates that the uniform single-beam model is a formal linear approximation. A physical suspension would require the actual leaf-spring stack, support geometry, preload, and available travel.

For the worn damper, the stiffness and mass do not change. Consequently, I , EI , k , ω_n , f_n , and c_{cr} remain unchanged. Set $\zeta_W = 0.10$ and calculate the new damping and decay rate as

$$c_W = \zeta_W c_{\text{cr}} = 211.863 \text{ N s/m}, \quad \alpha_W = \zeta_W \omega_n = 0.231797 \text{ s}^{-1}.$$

The smaller damping ratio gives

$$\omega_{d,W} = \omega_n \sqrt{1 - \zeta_W^2} = 2.30635 \text{ rad/s}, \quad f_{d,W} = 0.367068 \text{ Hz},$$

$$\tau_{d,W} = \frac{2\pi}{\omega_{d,W}} = 2.72429 \text{ s}.$$

The worn-damper roots and exponential constants are

$$s_{1,2}^W = -0.231797 \pm i 2.30635 \text{ s}^{-1}, \quad C_1^W = 62.500 - 6.281i \text{ mm}, \quad C_2^W = 62.500 + 6.281i \text{ mm}.$$

The corresponding real coefficients are

$$A_W = 125.0 \text{ mm}, \quad B_W = 12.563 \text{ mm}, \quad C_W = 125.630 \text{ mm}, \quad \phi_W = 1.47063 \text{ rad}.$$

Thus, the worn-damper response is

$$u_W(t) = 125.630e^{-0.231797t} \sin(2.30635t + 1.47063) \text{ mm}.$$

The response after one period and the final permanent 2% crossing are

$$t_{1,W} = \tau_{d,W} = 2.72429 \text{ s}, \quad u_{1,W} = u_0 e^{-\alpha_W t_{1,W}} = 66.475 \text{ mm},$$

$$t_{s,W} = 16.559 \text{ s}, \quad u_W(t_{s,W}) = +u_{0.02} = 2.50 \text{ mm}.$$

$c_{cr} = 2118.63 \text{ N s/m},$	$c_W = 211.86 \text{ N s/m},$	(B.4.16.a)
$\omega_{d,W} = 2.306 \text{ rad/s},$	$f_{d,W} = 0.3671 \text{ Hz},$	
$\tau_{d,W} = 2.724 \text{ s},$	$s_{1,2}^W = -0.2318 \pm i 2.306 \text{ s}^{-1},$	
$C_1^W = 62.50 - 6.281i \text{ mm},$	$C_2^W = 62.50 + 6.281i \text{ mm},$	
$A_W = 125.0 \text{ mm},$	$B_W = 12.56 \text{ mm},$	
$C_W = 125.6 \text{ mm},$	$\phi_W = 1.471 \text{ rad},$	
$t_{1,W} = 2.724 \text{ s},$	$u_{1,W} = 66.48 \text{ mm}$	

B.4.16.b. Comparison. The worn damper has a smaller decay rate, so its oscillations persist longer. Its damped frequency is slightly higher and its period is slightly shorter because reducing damping moves ω_d closer to ω_n . The one-period displacement increases from 24.68 mm to 66.48 mm, and the permanent 2% settling time increases from 6.09 s to 16.56 s. Both cases have the same initial maximum because they have the same initial displacement.

B4 worn. Displacement release

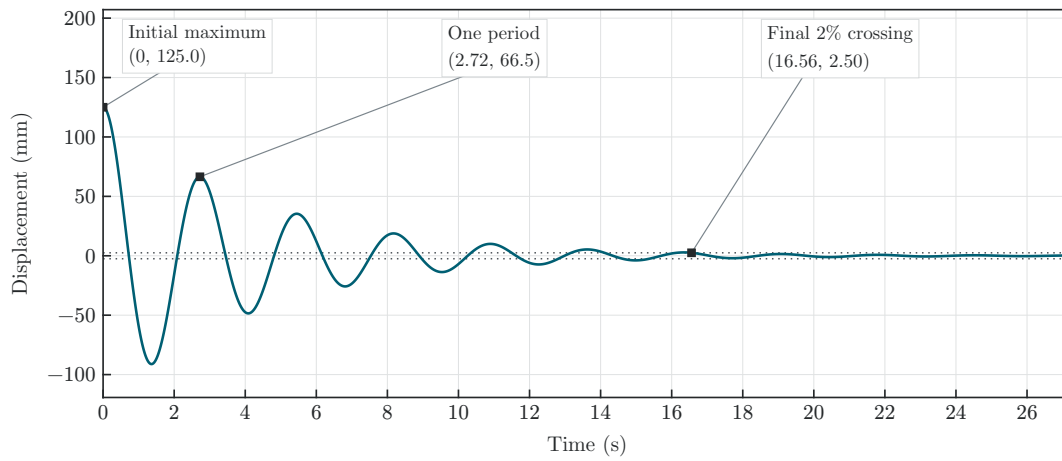


Figure 17: *B.4.16.a. Worn damper response for ten oscillations.*

B4. Worn damping increases the duration of motion

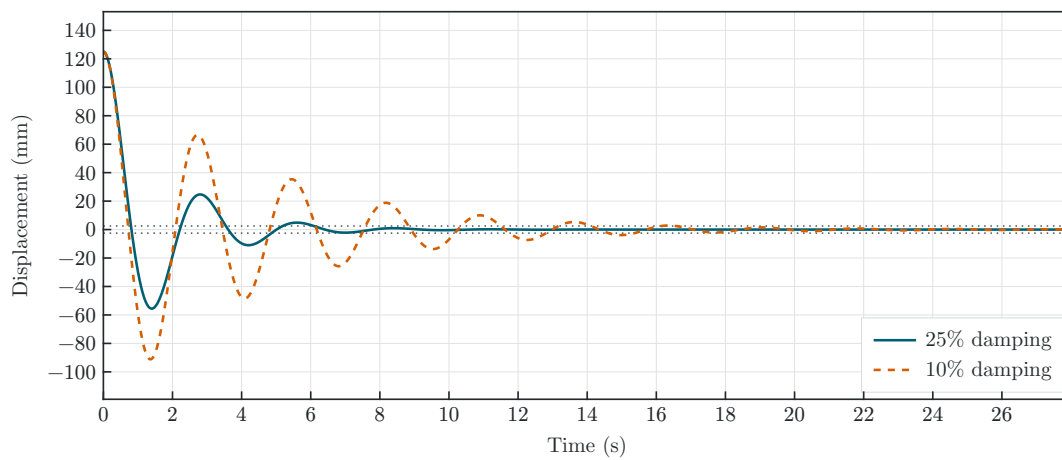


Figure 18: *B.4.16.b. Comparison of the original and worn-damper responses.*

MATLAB code. *MATLAB code used to reproduce the B.4.14 and B.4.16 response plots.*

```

01 L = .830; b = .065; h = .003; E = 200e9; m = 457; u0 = 125;
02 Nosc = 10; zeta = [.25 .10]; ts = [6.090194 16.558987];
03 I = b*h^3/12; k = 48*E*I/L^3; wn = sqrt(k/m);
04
05 for j = 1:2
06     wd = wn*sqrt(1-zeta(j)^2); tau = 2*pi/wd;
07     t = linspace(0,Nosc*tau,2000);
08     u = u0*exp(-zeta(j)*wn*t).*(cos(wd*t) + ...
09         zeta(j)*wn/wd*sin(wd*t));
10     subplot(2,1,j); plot(t,u,'Color',[0 .427 .459]); hold on
11     plot([0 t(end)], [.02*u0 .02*u0],':','Color',[.776 .365 0]);
12     plot([0 t(end)], [-.02*u0 -.02*u0],':','Color',[.776 .365 0]);
13     plot([0 tau ts(j)], [u0 u0*exp(-zeta(j)*wn*tau) .02*u0],'ks');
14     xlabel('Time (s)'); ylabel('Displacement (mm)'); grid on
15 end

```

B.5 Overdamped Spring-Mass-Damper Systems

Consider a spring-mass-damper system with natural frequency $f_n = 6.25 \text{ Hz}$ and various values of the damping ratio ζ . The system is given an initial displacement $u_0 = 13 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the rad/s frequency ω_n .
- (3) Assume overdamped conditions $\zeta > 1$ and develop the solution as follows:
 - (a) Write the expressions of the roots s_1, s_2 in terms of ω_n and ζ .
 - (b) Write the solution $u(t)$ in terms of s_1, s_2 and constants C_1, C_2 .
 - (c) Write the expression of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Take $\zeta = 2.3$ and calculate the numerical values of s_1, s_2, C_1 , and C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3 \text{ s}$.
- (4) Assume critically damped conditions $\zeta = 1$ and complete items (a)–(e).
 - (a) Write the expressions of the two partial solutions $u_1(t), u_2(t)$ in terms of ω_n .
 - (b) Write the expression of the complete solution in terms of ω_n and C_1, C_2 .
 - (c) Write the expressions of the constants C_1, C_2 in terms of initial displacement u_0 and initial velocity $\dot{u}_0$.
 - (d) Calculate the numerical values of C_1, C_2 .
 - (e) Plot the response $u(t)$ for $t = 0, \dots, 3 \text{ s}$.
- (5) Graduate work. Derive the ODE for $\zeta = 1$ and verify that the two solutions $u_1(t)$ and $u_2(t)$ from (4)(a) satisfy the ODE.
- (6) Graduate work. Some race-car manufacturers design for critical damping, whereas others use an overdamped design. Discuss the advantages and disadvantages of the two options.

Solution to B.5

The given initial conditions are $u(0) = u_0 = 13 \text{ mm}$ and $\dot{u}(0) = \dot{u}_0 = 0$. The natural frequency is specified by $f_n = 6.25 \text{ Hz}$.

B.5.1. Input data.

Quantity	Value	Unit
f_n	6.25	Hz
u_0	13.0	mm
$\dot{u}_0$	0	mm/s
t	0–3	s

B.5.2. Natural frequency.

The frequency in hertz is converted to angular frequency by multiplying by 2π .

$$\omega_n = 2\pi f_n = 2\pi(6.25) = 39.2699 \text{ rad/s.}$$

$$\boxed{\omega_n = 39.27 \text{ rad/s}} \quad (\text{B.5.2})$$

B.5.3. Overdamped response with $\zeta = 2.3$.

For free vibration, the characteristic equation is

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0.$$

(a) **Roots.** Since $\zeta > 1$, the discriminant is positive and the two roots are real. Define s_1 as the faster root and s_2 as the slower root.

$$s_1 = -\left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n, \quad s_2 = -\left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n.$$

$$\boxed{s_1 = -\left(\zeta + \sqrt{\zeta^2 - 1}\right)\omega_n, \quad s_2 = -\left(\zeta - \sqrt{\zeta^2 - 1}\right)\omega_n} \quad (\text{B.5.3.a})$$

(b) **General solution.** Each real root produces an exponential partial solution. Their linear combination is

$$\boxed{u(t) = C_1 e^{s_1 t} + C_2 e^{s_2 t}} \quad (\text{B.5.3.b})$$

(c) **Initial-condition constants.** Evaluating the response at $t = 0$ gives $C_1 + C_2 = u_0$. Differentiating first and then evaluating at $t = 0$ gives $s_1 C_1 + s_2 C_2 = \dot{u}_0$. Solving these two equations yields

$$\boxed{C_1 = \frac{\dot{u}_0 - s_2 u_0}{s_1 - s_2}, \quad C_2 = \frac{s_1 u_0 - \dot{u}_0}{s_1 - s_2}} \quad (\text{B.5.3.c})$$

(d) **Numerical values.** For $\zeta = 2.3$,

$$\sqrt{\zeta^2 - 1} = \sqrt{2.3^2 - 1} = 2.07123.$$

Substitution into the root expressions gives

$$\begin{aligned} s_1 &= -(2.3 + 2.07123)(39.2699) = -171.658 \text{ s}^{-1}, \\ s_2 &= -(2.3 - 2.07123)(39.2699) = -8.98372 \text{ s}^{-1}. \end{aligned}$$

Using $u_0 = 13 \text{ mm}$ and $\dot{u}_0 = 0$ in the constant expressions gives

$$C_1 = \frac{0 - (-8.98372)(13)}{-171.658 - (-8.98372)} = -0.71793 \text{ mm},$$

$$C_2 = \frac{(-171.658)(13) - 0}{-171.658 - (-8.98372)} = 13.71793 \text{ mm}.$$

$$\boxed{\begin{aligned} s_1 &= -171.658 \text{ s}^{-1}, & s_2 &= -8.98372 \text{ s}^{-1}, \\ C_1 &= -0.71793 \text{ mm}, & C_2 &= 13.71793 \text{ mm} \end{aligned}} \quad (\text{B.5.3.d})$$

Therefore, the overdamped displacement is

$$u_{\text{OD}}(t) = -0.71793e^{-171.658t} + 13.71793e^{-8.98372t} \text{ mm}.$$

(e) **Overdamped plot.** The response is plotted over the required interval $0 \leq t \leq 3 \text{ s}$.

B5. Critical and overdamped displacement release

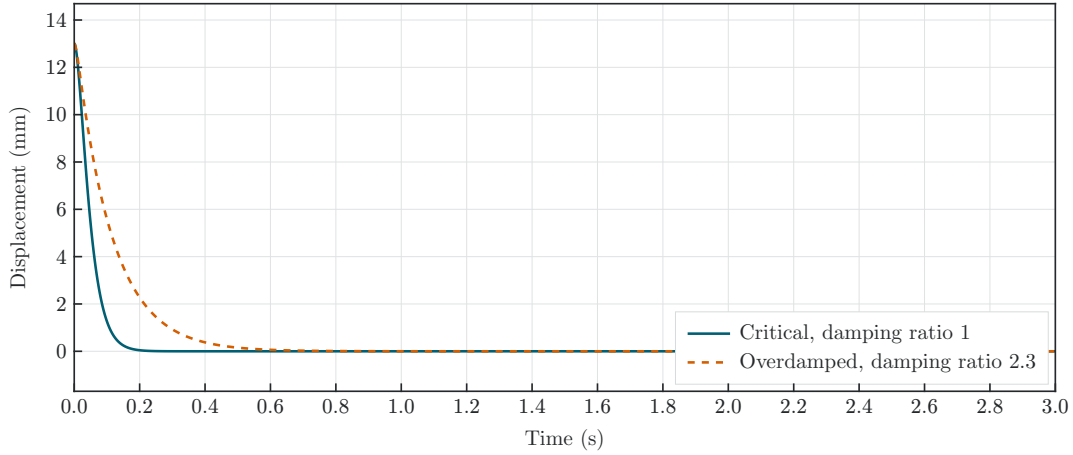


Figure 19: B.5.3.e. Overdamped and critical responses over $0 \leq t \leq 3$ s.

B.5.4. Critically damped response with $\zeta = 1$.

(a) **Partial solutions.** At $\zeta = 1$, the two roots merge into the repeated root $s = -\omega_n$. One solution is the exponential associated with that root. A second independent solution is obtained by multiplying the exponential by t .

$$\boxed{u_1(t) = e^{-\omega_n t}, \quad u_2(t) = t e^{-\omega_n t}} \quad (\text{B.5.4.a})$$

(b) **Complete solution.** The complete response is the linear combination of the two independent partial solutions.

$$\boxed{u(t) = (C_1 + C_2 t) e^{-\omega_n t}} \quad (\text{B.5.4.b})$$

(c) **Initial-condition constants.** At $t = 0$, the displacement condition gives $C_1 = u_0$. Differentiating the complete solution gives

$$\dot{u}(t) = [C_2 - \omega_n(C_1 + C_2 t)] e^{-\omega_n t}.$$

Evaluating at $t = 0$ gives $\dot{u}_0 = C_2 - \omega_n C_1$. Therefore,

$$\boxed{C_1 = u_0, \quad C_2 = \dot{u}_0 + \omega_n u_0} \quad (\text{B.5.4.c})$$

(d) **Numerical values.** The constants are

$$C_1 = 13.000 \text{ mm}, \quad C_2 = 0 + (39.2699)(13) = 510.509 \text{ mm/s}.$$

$$\boxed{C_1 = 13.00 \text{ mm}, \quad C_2 = 510.51 \text{ mm/s}} \quad (\text{B.5.4.d})$$

The critically damped displacement is consequently

$$u_{\text{CD}}(t) = (13 + 510.509t) e^{-39.2699t} \text{ mm.}$$

(e) Critically damped plot. The shorter time view makes the difference between the two nonoscillatory responses visible near the initial displacement.

B5. Critical and overdamped displacement release

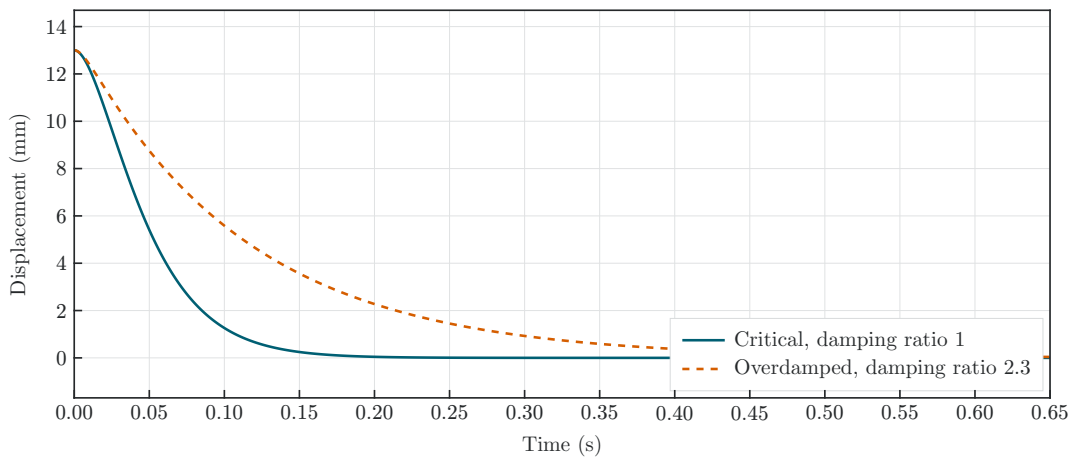


Figure 20: B.5.4.e. Detailed comparison of the critically damped and overdamped responses.

B.5.5. Verification of the critical-damping ODE.

At critical damping, $c = 2m\omega_n$ and $k = m\omega_n^2$. Substitution into $m\ddot{u} + c\dot{u} + ku = 0$, followed by division by m , gives

$$\ddot{u} + 2\omega_n\dot{u} + \omega_n^2u = 0.$$

For $u_1 = e^{-\omega_n t}$, differentiation gives $\dot{u}_1 = -\omega_n u_1$ and $\ddot{u}_1 = \omega_n^2 u_1$. Substitution gives

$$\ddot{u}_1 + 2\omega_n\dot{u}_1 + \omega_n^2u_1 = \omega_n^2u_1 - 2\omega_n^2u_1 + \omega_n^2u_1 = 0.$$

For $u_2 = te^{-\omega_n t}$, the first two derivatives are

$$\dot{u}_2 = (1 - \omega_n t)e^{-\omega_n t}, \quad \ddot{u}_2 = (\omega_n^2 t - 2\omega_n)e^{-\omega_n t}.$$

Substitution into the ODE gives

$$\begin{aligned}\ddot{u}_2 + 2\omega_n\dot{u}_2 + \omega_n^2 u_2 &= (\omega_n^2 t - 2\omega_n)e^{-\omega_n t} + 2\omega_n(1 - \omega_n t)e^{-\omega_n t} + \omega_n^2 t e^{-\omega_n t} \\ &= 0.\end{aligned}$$

Both partial solutions satisfy the critical-damping ODE, so their linear combination is also a solution.

B.5.6. Critical versus overdamped design.

Both designs return to equilibrium without oscillation. Critical damping has the repeated decay rate $-\omega_n$, so it gives the fastest nonoscillatory return for the specified ω_n and initial conditions. Its main disadvantage is that the damping must be tuned close to the critical value. Changes in mass, stiffness, or damper properties move the system away from that condition.

For $\zeta = 2.3$, the overdamped response contains a fast rate of -171.658 s^{-1} and a slow rate of -8.98372 s^{-1} . The fast term reduces the displacement quickly at first, but the slow term controls the long tail and makes the final settling slower. Overdamping can therefore provide more tolerance against crossing into oscillatory behavior, but it sacrifices response speed and can increase damper forces during rapid motion. A race-car suspension must balance these effects with tire contact, road inputs, suspension travel, and damper-force limits.

MATLAB code. *MATLAB script for the B.5 overdamped and critically damped response plots.*

```
01 fn = 6.25; u0 = 13; v0 = 0; z = 2.3;
02 t = linspace(0,3,2000); wn = 2*pi*fn;
03
04 q = sqrt(z^2-1); s1 = -(z+q)*wn; s2 = -(z-q)*wn;
05 C1 = (v0-s2*u0)/(s1-s2); C2 = (s1*u0-v0)/(s1-s2);
06 uOD = C1*exp(s1*t)+C2*exp(s2*t);
07 uCD = (u0+(v0+wn*u0)*t).*exp(-wn*t);
08
09 plot(t,uCD,'Color',[0 .427 .459],'LineWidth',1.5); hold on
10 plot(t,uOD,'--','Color',[.776 .365 0],'LineWidth',1.5)
11 xlabel('Time (s)'); ylabel('Displacement (mm)'); grid on
12 legend('Critical, zeta = 1','Overdamped, zeta = 2.3')
```

B.6 Analyze Spring-Mass-Damper Forced Vibration

Consider a mass-spring-damper vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 21.

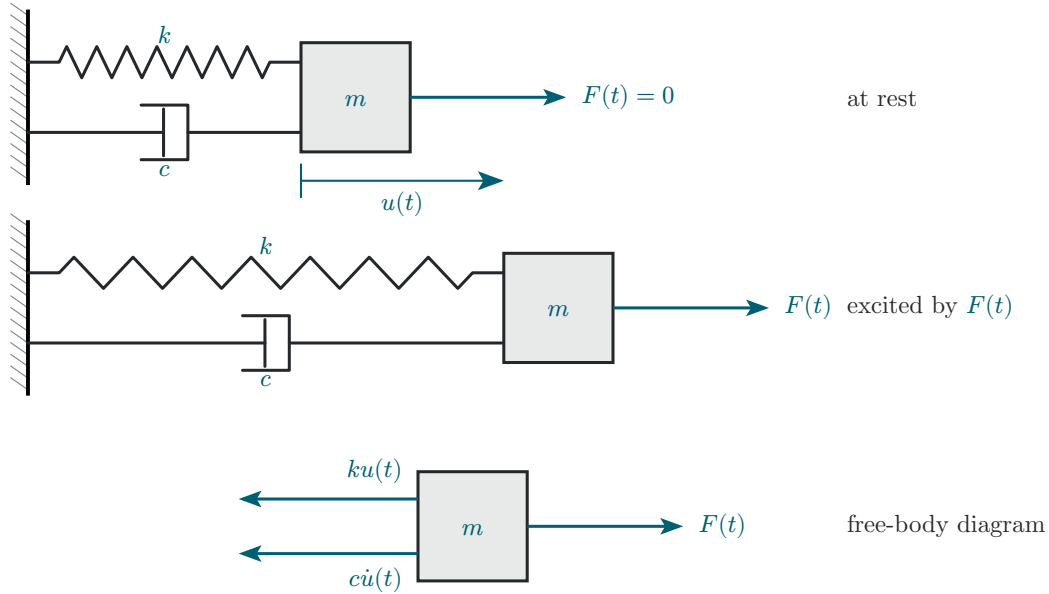


Figure 21: Spring-mass-damper system excited by force $F(t)$.

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) as an ODE in terms of m , c , k , $F(t)$.
- (5) Assume harmonic excitation $F(t) = \hat{F} e^{i\omega t}$, substitute in (4), and divide by m .
- (6) Define the normalized excitation amplitude $\hat{f} = \hat{F}/m$ and recall the definitions of ω_n and ζ to obtain a nonhomogeneous linear ODE in ω_n , ζ , $\hat{f}$.
- (7) Write the general solution $u(t)$ as a sum of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.
- (8) Write the general form of the complementary solution $u_c(t)$.
- (9) Find the particular solution $u_p(t)$.
- (10) Use (8), (9) to write the complete solution $u(t)$; discuss the meaning of transient response and steady-state response.
- (11) Find an expression for the steady-state response amplitude $\hat{u}_{ss}$ in terms of ω_n , ζ , $\hat{f}$, and ω .
- (12) Develop the expression for the frequency response function $\text{FRF}(\omega)$ in terms of ω_n , ζ , ω .
- (13) Discuss the variation with frequency of FRF magnitude and phase.

- (14) Give the definition of the magnification factor $M(\omega)$.
- (15) Give the expression for the magnification factor $M(\omega)$ in terms of ω_n and ζ .
- (16) Give the definition of the resonance frequency ω_r .
- (17) Give the expression for the resonance frequency ω_r in terms of ω_n and ζ .
- (18) Give the expression of the peak magnification M_r .
- (19) Give the expression for the FRF phase $\phi(p)$, where $p = \omega/\omega_n$.
- (20) Give the definition of phase resonance and deduce the FRF expression at phase resonance.
- (21) Calculate the magnification factor at phase resonance.
- (22) Calculate the magnitude of the physical response at phase resonance.
- (23) Discuss the difference between true resonance and phase resonance.
- (24) Write and compare the expressions for the resonance frequency ω_r and damped frequency ω_d .
- (25) Establish a formula to calculate the RPM corresponding to a rad/sec frequency ω .

Solution to B.6

B.6.1. Assumptions. The model has one translational degree of freedom $u(t)$, a point mass m , a linear spring with stiffness k , and a viscous damper with coefficient c . The support is fixed, the motion is small, and the applied force $F(t)$ acts in the positive x direction. Spring and damper forces oppose the motion, and the parameters m , c , and k remain constant.

B.6.2. Free-body diagram and force sum. Take positive x to the right. The applied force is $F(t)$, while the spring and damper reactions act to the left. Therefore,

$$\sum F_x = F(t) - c\dot{u}(t) - ku(t) \quad (\text{B.6.2})$$

The corresponding isolated mass is shown below.

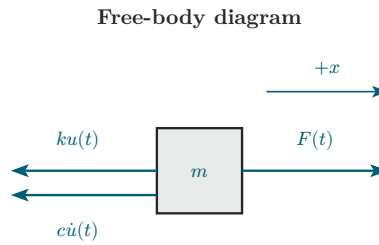


Figure 22: B.6.2. Free-body diagram for the forced spring-mass-damper system.

B.6.3. Newton's law of motion. Newton's law relates the net force to the mass acceleration.

$$\sum F_x = m\ddot{u}(t) \quad (\text{B.6.3})$$

B.6.4. Equation of motion. Equating the force sum to the inertial term gives

$$F(t) - c\dot{u}(t) - ku(t) = m\ddot{u}(t).$$

Moving the spring and damper terms to the left gives the governing ODE.

$$m\ddot{u}(t) + c\dot{u}(t) + ku(t) = F(t) \quad (\text{B.6.4})$$

B.6.5. Harmonic excitation and division by mass. Let the applied force be represented by a complex harmonic function,

$$F(t) = \hat{F}e^{i\omega t}.$$

Substitution into the equation of motion and division by m gives

$$\ddot{u} + \frac{c}{m}\dot{u} + \frac{k}{m}u = \frac{\hat{F}}{m}e^{i\omega t} \quad (\text{B.6.5})$$

The physical force and displacement are obtained by taking the real parts of the complex expressions.

B.6.6. Normalized nonhomogeneous ODE. Define the normalized force amplitude and the natural frequency by

$$\hat{f} = \frac{\hat{F}}{m}, \quad \omega_n = \sqrt{\frac{k}{m}}.$$

The damping ratio is

$$\zeta = \frac{c}{2\sqrt{mk}} = \frac{c}{2m\omega_n}, \quad \frac{c}{m} = 2\zeta\omega_n, \quad \frac{k}{m} = \omega_n^2.$$

Substitution into the divided equation produces the normalized forced-vibration ODE.

$$\ddot{u} + 2\zeta\omega_n\dot{u} + \omega_n^2u = \hat{f}e^{i\omega t} \quad (\text{B.6.6})$$

B.6.7. General solution. The right-hand side is nonzero, so the solution is the sum of a complementary solution and a particular solution.

$$u(t) = u_c(t) + u_p(t) \quad (\text{B.6.7})$$

The complementary part satisfies the homogeneous equation, and the particular part accounts for the harmonic force.

B.6.8. Complementary solution. For the underdamped case $0 \leq \zeta < 1$, define the damped frequency as

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}.$$

The homogeneous equation then has the real solution

$$u_c(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \quad (\text{B.6.8})$$

where C and ϕ are determined by the initial conditions after the particular solution is included. Equivalently, the same response can be written as

$$u_c(t) = e^{-\zeta\omega_n t} [A \cos(\omega_d t) + B \sin(\omega_d t)].$$

B.6.9. Particular solution. Because the excitation is proportional to $e^{i\omega t}$, seek a particular solution in the same form,

$$u_p(t) = \hat{u}_p e^{i\omega t}.$$

Differentiation gives $\dot{u}_p = i\omega \hat{u}_p e^{i\omega t}$ and $\ddot{u}_p = -\omega^2 \hat{u}_p e^{i\omega t}$. Substitution into the normalized ODE gives

$$(-\omega^2 + 2i\zeta\omega_n\omega + \omega_n^2) \hat{u}_p = \hat{f}.$$

Solving for the complex displacement amplitude gives

$$\hat{u}_p = \frac{\hat{f}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega}, \quad u_p(t) = \frac{\hat{f}e^{i\omega t}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \quad (\text{B.6.9})$$

B.6.10. Complete response and its meaning. Combining the complementary and particular parts gives

$$u(t) = Ce^{-\zeta\omega_n t} \sin(\omega_d t + \phi) + \frac{\hat{f}e^{i\omega t}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \quad (\text{B.6.10})$$

The first term is the transient response. It decays because of the factor $e^{-\zeta\omega_n t}$. The second term is the steady-state response. It remains after the transient has decayed and oscillates at the forcing frequency ω . Initial conditions must be applied to the complete response, since both terms contribute at the start of the motion.

B.6.11. Steady-state response amplitude. The complex amplitude is the coefficient multiplying $e^{i\omega t}$ in the particular response.

$$\hat{u}_{ss}(\omega) = \frac{\hat{f}}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega} \quad (\text{B.6.11})$$

Using $\hat{f} = \hat{F}/m$ gives the equivalent physical-parameter form

$$\hat{u}_{ss}(\omega) = \frac{\hat{F}}{k - m\omega^2 + i c \omega}.$$

B.6.12. Frequency response function. The quasi-static displacement is obtained by taking $\omega \rightarrow 0$.

$$u_{\text{qst}} = \frac{\hat{F}}{k} = \frac{\hat{f}}{\omega_n^2}.$$

Normalize the steady-state amplitude by this displacement to obtain

$$\boxed{\text{FRF}(\omega) = \frac{\hat{u}_{\text{ss}}}{u_{\text{qst}}} = \frac{\omega_n^2}{\omega_n^2 - \omega^2 + 2i\zeta\omega_n\omega}} \quad (\text{B.6.12})$$

With $p = \omega/\omega_n$, this becomes

$$\text{FRF}(p) = \frac{1}{1 - p^2 + 2i\zeta p}.$$

B.6.13. FRF variation with frequency. At zero excitation frequency, $\text{FRF}(0) = 1$, so the response equals the quasi-static displacement and has zero phase lag. As p increases, the magnitude can rise near the natural-frequency range and then decreases toward zero. The phase is negative because the displacement lags the applied force. It changes from 0 at low frequency toward $-\pi$ as p becomes large. Increasing ζ lowers and broadens the magnitude peak. When $\zeta \geq 1/\sqrt{2}$, the magnitude has no nonzero interior peak.

B.6.14. Magnification-factor definition. The magnification factor is the magnitude of the normalized steady-state response.

$$\boxed{M(\omega) = |\text{FRF}(\omega)| = \frac{|\hat{u}_{\text{ss}}|}{u_{\text{qst}}}} \quad (\text{B.6.14})$$

B.6.15. Magnification factor in terms of ω_n and ζ . Taking the magnitude of the normalized FRF gives

$$\boxed{M(\omega) = \frac{\omega_n^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\zeta\omega_n\omega)^2}} = \frac{1}{\sqrt{(1 - p^2)^2 + (2\zeta p)^2}}} \quad (\text{B.6.15})$$

B.6.16. Resonance-frequency definition. The true resonance frequency is the frequency at which the magnification factor is maximum.

$$\boxed{\omega_r = \arg \max_{\omega \geq 0} M(\omega)} \quad (\text{B.6.16})$$

B.6.17. Resonance frequency expression. It is convenient to minimize the reciprocal squared magnitude,

$$M^{-2}(p) = Q(p) = (1 - p^2)^2 + 4\zeta^2 p^2.$$

Differentiation gives

$$\frac{dQ}{dp} = 4p (p^2 + 2\zeta^2 - 1).$$

The nonzero stationary point exists when $0 \leq \zeta < 1/\sqrt{2}$, which gives

$$\boxed{\omega_r = \omega_n \sqrt{1 - 2\zeta^2}, \quad 0 \leq \zeta < \frac{1}{\sqrt{2}}} \quad (\text{B.6.17})$$

For $\zeta \geq 1/\sqrt{2}$, the square root is not real and the maximum occurs at the zero-frequency limit rather than at a nonzero resonance frequency.

B.6.18. Peak magnification. Substitute $p_r^2 = 1 - 2\zeta^2$ into the magnification expression. The result is

$$\boxed{M_r = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}, \quad 0 < \zeta < \frac{1}{\sqrt{2}}} \quad (\text{B.6.18})$$

For zero damping, the ideal model has an unbounded response at $\omega = \omega_n$.

B.6.19. FRF phase. The phase is the argument of the complex FRF. Using the quadrant-aware inverse tangent gives

$$\boxed{\phi(p) = \arg[\text{FRF}(p)] = -\text{atan2}(2\zeta p, 1 - p^2)} \quad (\text{B.6.19})$$

This expression gives $\phi(0) = 0$ and approaches $-\pi$ as $p \rightarrow \infty$.

B.6.20. Phase resonance. Phase resonance is the condition that the displacement lags the applied force by 90° , or $\phi = -\pi/2$. The real part of the FRF denominator must then vanish,

$$1 - p^2 = 0.$$

For positive frequency, this gives $p = 1$ and $\omega = \omega_n$. At this frequency,

$$\boxed{\text{FRF}(\omega_n) = \frac{1}{2i\zeta} = -\frac{i}{2\zeta}} \quad (\text{B.6.20})$$

B.6.21. Magnification at phase resonance. Taking the magnitude of the phase-resonance FRF gives

$$\boxed{M_{90} = |\text{FRF}(\omega_n)| = \frac{1}{2\zeta}} \quad (\text{B.6.21})$$

B.6.22. Physical response at phase resonance. Multiplying the normalized FRF by the

quasi-static displacement gives the complex response amplitude,

$$\hat{u}_{90} = u_{\text{qst}} \text{FRF}(\omega_n) = -i \frac{\hat{F}}{2\zeta k}.$$

Therefore, the physical response magnitude is

$$|\hat{u}_{90}| = \frac{\hat{F}}{2\zeta k} = \frac{\hat{F}}{c\omega_n} \quad (\text{B.6.22})$$

B.6.23. True resonance versus phase resonance. True resonance identifies the frequency at which the response magnitude is maximum. For light damping, it occurs at $\omega_r = \omega_n \sqrt{1 - 2\zeta^2}$, which is below ω_n . Phase resonance identifies the frequency at which the response lags by 90° , and it occurs at $\omega = \omega_n$ for every positive damping ratio. These frequencies are equal only in the zero-damping limit. For $\zeta \geq 1/\sqrt{2}$, there is no nonzero true-resonance peak, but the phase-resonance condition still occurs at ω_n .

B.6.24. Resonance and damped frequencies. For an underdamped system,

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}.$$

When $0 < \zeta < 1/\sqrt{2}$, the three frequencies satisfy

$$\omega_r = \omega_n \sqrt{1 - 2\zeta^2} < \omega_d = \omega_n \sqrt{1 - \zeta^2} < \omega_n \quad (\text{B.6.24})$$

Thus, the true resonance frequency is below the damped free-vibration frequency, which is below the undamped natural frequency.

B.6.25. Conversion from rad/s to RPM. One complete excitation cycle per revolution corresponds to an angular speed of 2π radians per revolution. Therefore, a frequency ω in rad/s corresponds to

$$N_{\text{RPM}} = \frac{60\omega}{2\pi} = \frac{30\omega}{\pi} \text{ RPM} \quad (\text{B.6.25})$$

B.7 Calculate Flex-Beam Response to Slightly Unbalanced Electric Motor Excitation

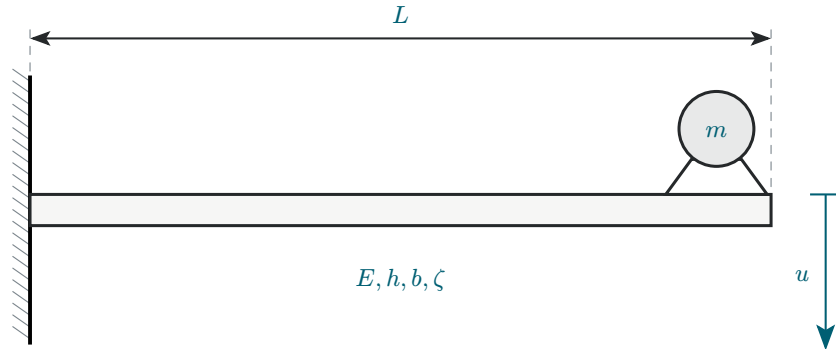


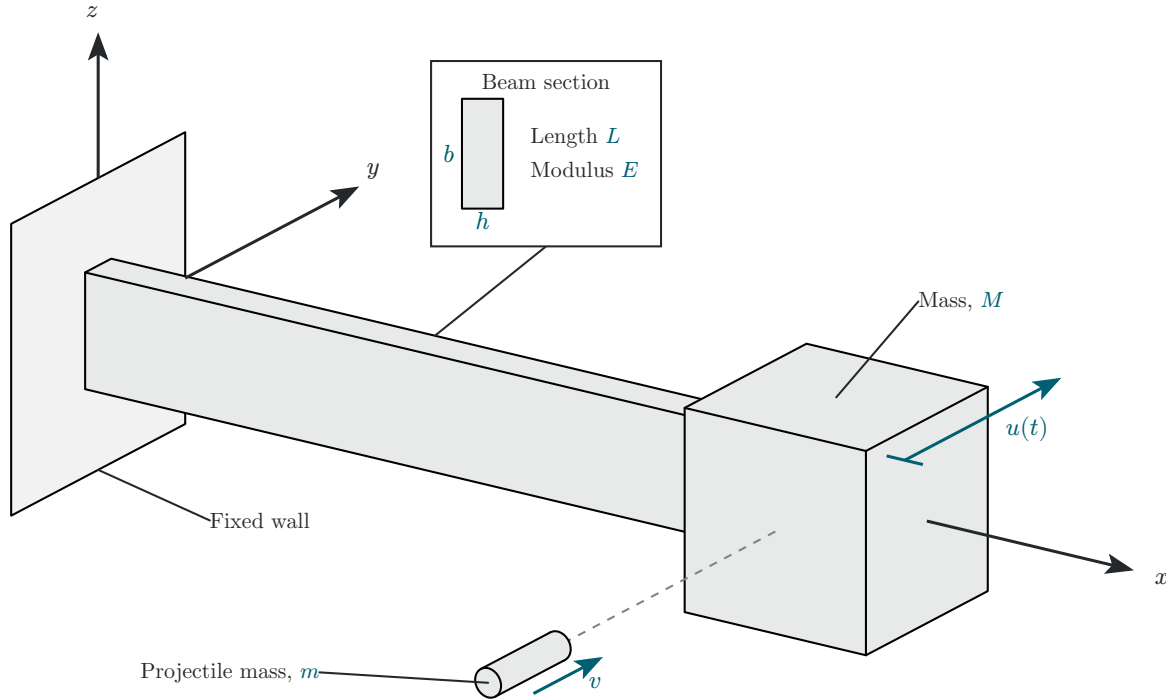
Figure 23: Damped cantilever beam excited by force $F(t)$ from slightly unbalanced electric motor.

Consider a cantilever beam of length $L = 568 \text{ mm}$, elastic modulus $E = 200 \text{ GPa}$, cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$, and damping ratio $\zeta = 14\%$. (The damping is achieved through a proprietary damping treatment applied on the beam surface.) An electric motor of mass $m = 0.52 \text{ kg}$ operating at $1,500 \text{ RPM}$ is attached at the beam tip as shown in Figure 23. The electric motor is slightly unbalanced, producing a slight vibratory excitation $F(t) = \hat{F}e^{i\omega t}$ with $\hat{F} = 12 \text{ mN}$. The excitation frequency ω depends on the electric motor speed.

Do the following.

- (1) Display input data.
- (2) Calculate:
 - (a) Cross-section second moment of area I , flexural stiffness EI , and spring constant k .
 - (b) Natural frequency ω_n rad/s, critical damping c_{cr} , physical damping c .
 - (c) Quasi-static deflection u_{qst} corresponding to $\omega_n \rightarrow 0$.
- (3) Calculate the values of:
 - (a) Resonant frequency ω_r and damped frequency ω_d .
 - (b) Peak magnification M_r .
 - (c) Magnification at phase resonance M_{90} .
 - (d) Physical response $\hat{u}_{90}$ at phase resonance and its magnitude $|\hat{u}_{90}|$.
 - (e) Calculate the Hz frequencies f_r , f_n and the corresponding RPM values.
- (4) Assume excitation at phase resonance:
 - (a) Calculate the excitation frequency ω in rad/s and f in Hz.
 - (b) Calculate period of oscillation τ .
 - (c) Plot the time response for $N_{osc} = 10$ oscillations. Hint: plot the real part of the complex displacement $u(t)$.
- (5) Graduate work. Complete items (a)–(d).

- (a) Calculate and display the operational frequency f_0 in Hz and the corresponding ω_0 in rad/sec. Plot the magnitude and phase of the frequency response function FRF vs. excitation f in Hz from quasi-static to f_0 with $N_f = 5000$ points.
- (b) Calculate and display the FRF magnitude at phase-resonance RPM, at double the phase resonance RPM, and at operational RPM. Display the corresponding Hz frequencies. Put the corresponding data tips on the magnitude plot.
- (c) Calculate the vibration magnitude $|u_0|$ when the electric motor runs at its operational RPM.
- (d) Comment on what vibration would be observed as the motor is switched on and accelerates to reach its operational RPM.

B.8 Graduate Work. Calculate Horizontal Flex-Beam Response to Impact**Figure 24:** *Impact excitation of a horizontal flex-beam vibration system.*

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $M = 53 \text{ kg}$ attached at its tip as shown in Figure 24. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The beam is initially at rest. A small projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. This impact imparts an initial velocity to the system that starts to oscillate with motion $u(t)$ contained in the horizontal plane.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as a result of the interaction between the projectile and the mass.
- (6) Calculate the rad/sec natural frequency ω_n and period of oscillation τ .
- (7) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response calculated at item (7).

- (9) Discuss your results.

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Vertical Spring-Mass Vibration System

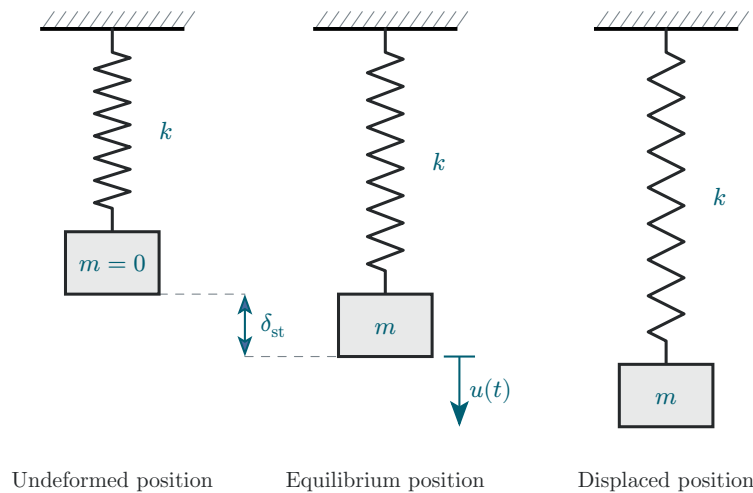


Figure 25: Vertical spring-mass vibration system.

Consider a vertical spring-mass vibration system consisting of a particle of mass m hanging on an elastic spring of stiffness k as shown in Figure 25.

Do the following.

- (1) Assume static equilibrium.
 - (a) Calculate the weight W acting on the mass m .
 - (b) Draw free-body diagram (FBD).
 - (c) Write equilibrium equation.
 - (d) Solve for the static displacement δ_{st} .
- (2) Assume vibration with time-varying vertical displacement $u(t)$.
 - (a) Draw free-body diagram (FBD).
 - (b) Deduce equation of motion (EOM).
 - (c) Compare the EOM deduced here with the EOM derived for horizontal spring-mass vibration system of Problem B.1.
- (3) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

C.2 Analyze Vertical Flex-Beam Vibration

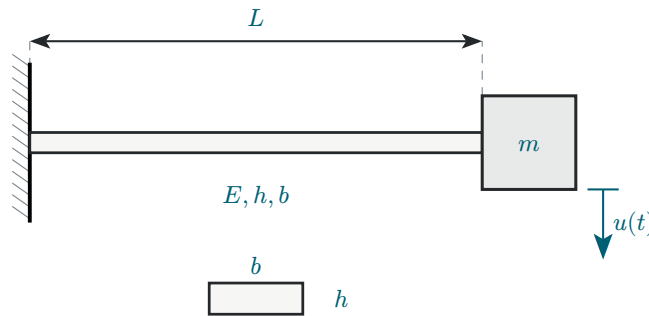


Figure 26: Vertical flex-beam vibrating system.

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 26. The beam cross-section properties are E , h , b .

Do the following.

- (1) Calculate the cross-sectional stiffness EI of the beam for vertical flexural deformation.
- (2) Calculate the spring constant k of the system.
- (3) Assume static equilibrium:
 - (a) Calculate the weight W acting on the mass m .
 - (b) Draw free-body diagram (FBD).
 - (c) Write equilibrium equation.
 - (d) Solve for the static displacement δ_{st} .
- (4) Assume vibration with time-varying vertical displacement $u(t)$ measured from the static equilibrium position:
 - (a) Draw free-body diagram (FBD).
 - (b) Deduce equation of motion (EOM).
 - (c) Compare the EOM deduced here with the EOM derived for horizontal flex-beam vibration system of Problem C.4.
- (5) Graduate work. Comment on how the EOM would change in the presence of an arbitrary force field acting at an angle θ with respect to the vertical axis.

C.3 Analyze Horizontal Flex-Beam Vibration

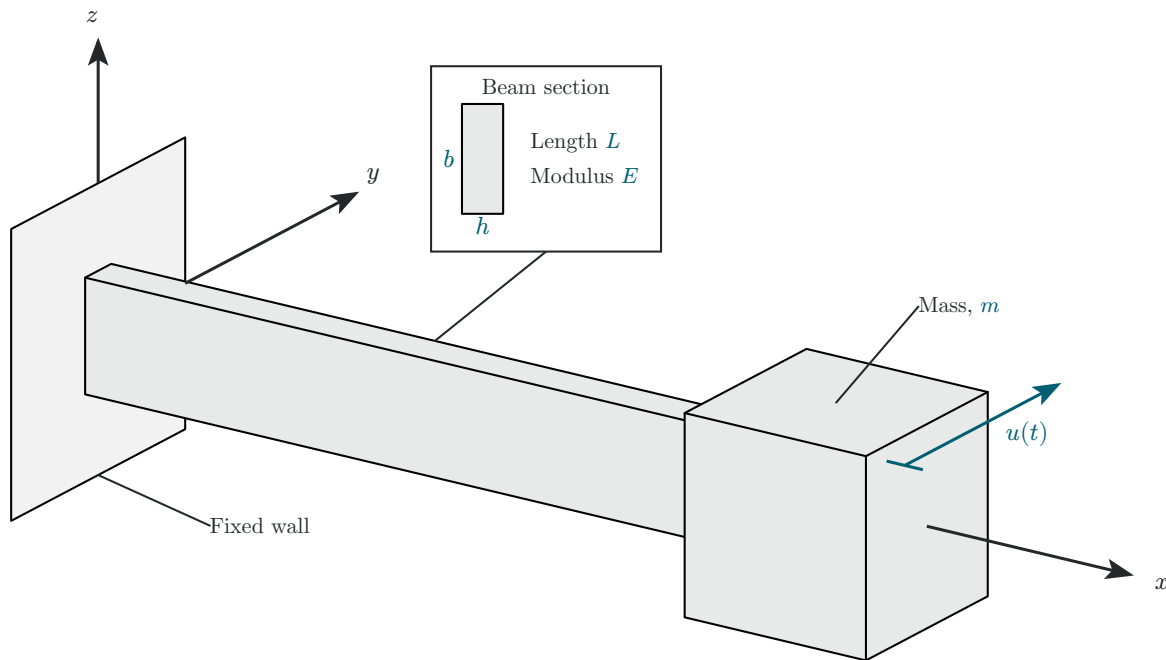


Figure 27: Horizontal flex-beam vibrating system.

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 27. The beam cross-section properties are E , h , b .

Do the following.

- (1) Assume vibration with horizontal time-varying displacement $u(t)$ as shown in Figure 27.
- (2) Calculate the cross-sectional stiffness EI of the beam for flexural deformation consistent with $u(t)$.
- (3) Calculate the flex-beam spring constant k .
- (4) Draw free-body diagram (FBD) including all the forces acting on the mass m .
- (5) Deduce equation of motion (EOM).
- (6) Compare the EOM deduced here with the EOM derived in Problem B.1.
- (7) Solve the initial value problem (IVP) and find the response for simultaneous application of both initial displacement u_0 and initial velocity $\dot{u}_0$.

C.4 Calculate Horizontal Flex-Beam Response to Initial Displacement

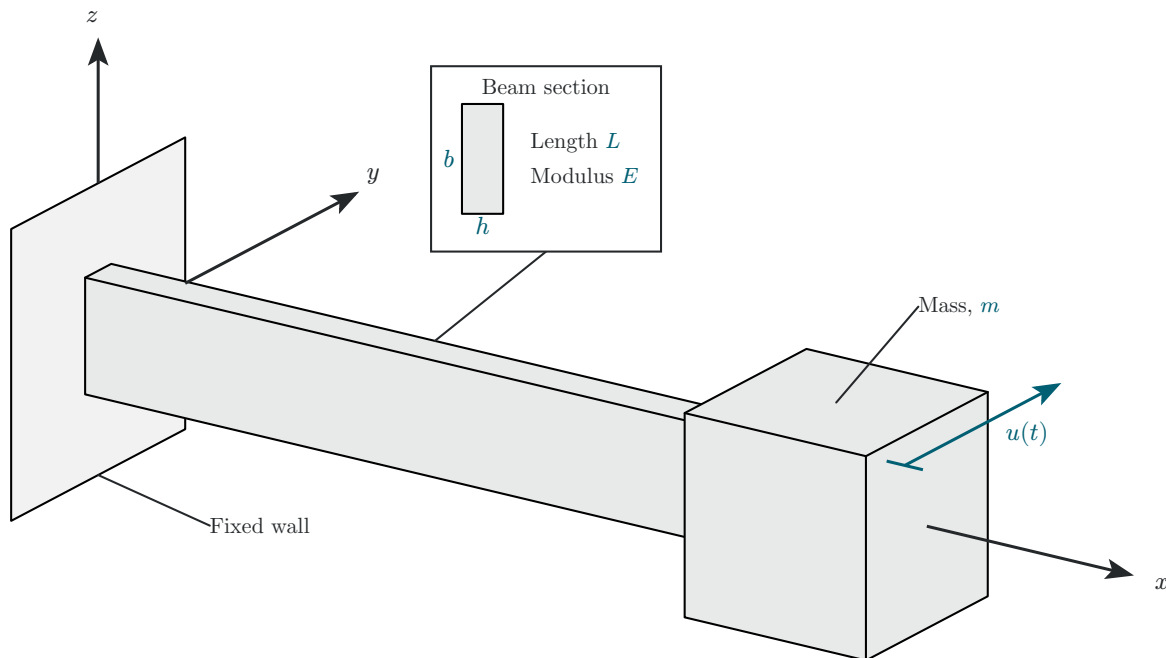


Figure 28: *Flex-beam vibrating system.*

Consider a cantilever beam of length $L = 568 \text{ mm}$ with a mass $m = 5.35 \text{ kg}$ attached at its tip as shown in Figure 28. The beam material is steel with $E = 200 \text{ GPa}$. The beam cross-section dimensions are $h = 3 \text{ mm}$, $b = 37 \text{ mm}$. The tip mass is given an initial displacement $u_0 = 5.4 \text{ mm}$ with zero initial velocity.

Do the following.

- (1) Display input data.
- (2) Calculate the second moment of area I for flexing consistent with the direction of motion $u(t)$.
- (3) Calculate the cross-sectional stiffness EI of the beam.
- (4) Calculate the spring constant k of the system.
- (5) Calculate the natural frequency in rad/sec ω_n .
- (6) Calculate the natural frequency in Hz f_n .
- (7) Calculate the period of oscillation in sec τ .
- (8) Calculate the position of the mass $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (9) Plot the response of the system for $N_{\text{osc}} = 25$ oscillations and use a data tip to find $u(t_1)$ at t_1 .
- (10) Discuss your results.

C.5 Analyze Spring-Mass Forced Vibration

Consider a spring-mass vibrating system excited by a time-dependent force $F(t)$ as shown in Figure 29.

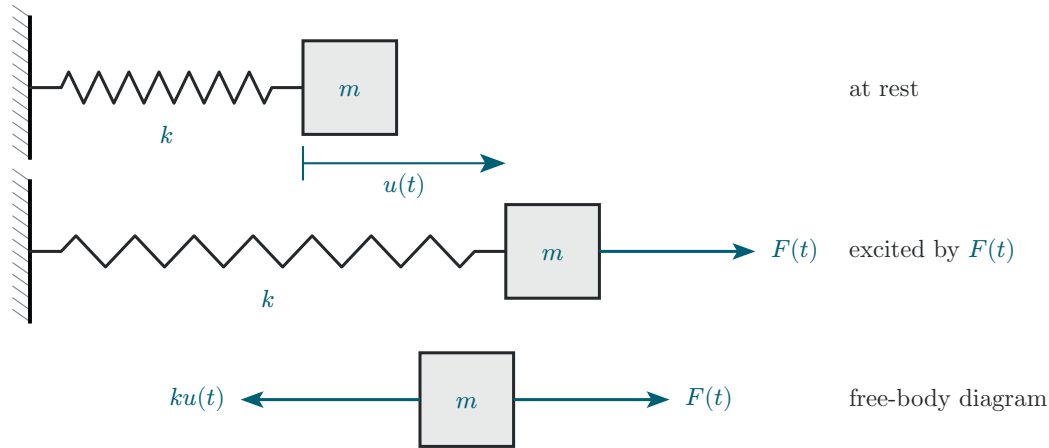


Figure 29: Spring-mass system excited by force $F(t)$.

General Analysis of Spring-Mass Forced Vibration

Do the following.

- (1) State your assumptions in performing this analysis.
- (2) Draw free-body diagram (FBD) and deduce the sum of forces in the x direction.
- (3) Write NLM equation.
- (4) Deduce the equation of motion (EOM) and express it as a nonhomogeneous ODE in terms of m , k , $F(t)$.
- (5) Write the ODE general solution as the superposition of a complementary solution $u_c(t)$ and a particular solution $u_p(t)$.
- (6) Recall the complementary solution $u_c(t)$ as the solution of the homogeneous equation derived in Problem B.1 and write it in terms of trigonometric functions multiplied by constants A and B .

Spring-Mass Response to a Suddenly Applied Load

- (7) Assume the loading $F(t)$ is a suddenly applied load of magnitude F_0 .
- (8) Find the particular solution for this case of a suddenly applied load.
- (9) Write the complete solution as the sum of $u_c(t)$ from (6) and $u_p(t)$ from (8).
- (10) Assume “at rest” initial conditions to find the constants A and B and write the complete solution $u(t)$ as a function of F_0 , k , and $\cos \omega_n t$.
- (11) Use (10) to find the largest displacement $u_{\max}$ and compare it with the static displacement u_{st} .

C.6 Calculate Cantilever Response to Suddenly Applied Weight

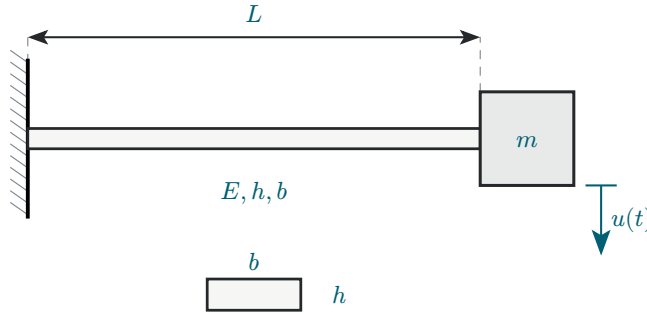


Figure 30: Cantilever beam with tip mass.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a cantilever beam of length $L = 568 \text{ mm}$ and a mass $m = 5.35 \text{ kg}$ that has to be attached to its tip as shown in Figure 30. The beam is made of steel with elastic modulus $E = 200 \text{ GPa}$, yield stress $Y = 710 \text{ MPa}$, and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$.

Do the following.

- (1) Display input data.
- (2) Calculate the cross-section second moment of area I , the flexural stiffness EI , the spring constant k , the natural frequency ω_n rad/s, f_n Hz, and period of oscillation τ of the system.
- (3) Calculate the weight F_0 of the mass m .
- (4) Static case. Assume the mass is attached to the undeformed beam and then released gradually without inducing vibration. Calculate the static displacement u_{st} .
- (5) Dynamic case. Assume the mass is attached to the undeformed beam and then released suddenly, inducing vibration.
 - (a) Calculate the maximum dynamic displacement u_{dyn} .
 - (b) Plot the response $u(t)$ measured from the undeformed position for $N_{\text{osc}} = 10$ oscillations.
 - (c) Place a data tip on the plot to identify the maximum displacement and compare it with the result of item (5)(a).
- (6) Graduate work. Calculate the maximum stress at the root of the beam and the safety factor $\text{SF} = Y/\sigma$, and comment on your results for the following two cases.
 - (a) Static-case stress σ_{st} .
 - (b) Dynamic-case stress σ_{dyn} .
- (7) Discuss your results.

C.7 Analyze Flex-Beam Pendulum

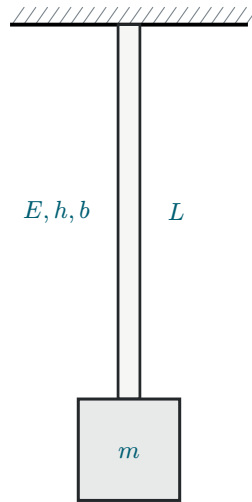


Figure 31: *Flex-beam pendulum.*

Consider a cantilever beam of length L with a mass m attached at its tip as shown in Figure 31. The beam is fixed into the ceiling and is hanging downward along the z axis. The mass may oscillate sideways in either x or y directions. The beam cross-section properties are E, h, b .

Do the following.

- (1) Calculate the minor and major second moments of area I_1 and I_2 .
- (2) Assume a horizontal displacement $u(t)$ which produces flexure about the minor I_1 .
- (3) Draw free-body diagram (FBD).
- (4) Write Newton's law of motion (NLM) equation.
- (5) Apply small-angle approximation and deduce the equation of motion (EOM). Clearly state your assumptions.
- (6) Define the natural frequency $\omega_{n,\text{FBP}}$ of the flex-beam pendulum.
- (7) Deduce the governing equation.
- (8) Compare your results with those of Problem B.1 and write the general solution.
- (9) Solve the initial value problem (IVP), i.e., deduce the response for simultaneous application of initial displacement u_0 and initial velocity $\dot{u}_0$.
- (10) Graduate work. Explain how the solution would change if flexure were about I_2 .

C.8 Calculate Flex-Beam Pendulum Response to Impact

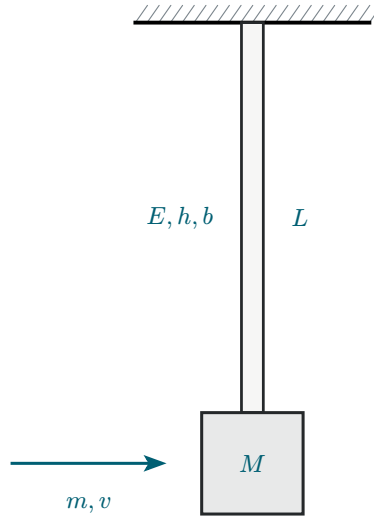


Figure 32: Impact excitation of a vertical flex-beam pendulum.

Take $g = 9.81 \text{ m/s}^2$ as the generally accepted value of gravitational acceleration. Consider a vertical flex-beam pendulum of length $L = 568 \text{ mm}$ with a mass $M = 42 \text{ kg}$ attached at its end, as shown in Figure 32. The beam is steel with $E = 200 \text{ GPa}$ and cross-section dimensions $h = 3 \text{ mm}$ and $b = 37 \text{ mm}$. The beam is initially at rest. A projectile of mass $m = 18 \text{ g}$ and velocity $v = 935 \text{ m/s}$ hits the mass and remains embedded in it. The projectile orientation causes the beam to flex about its minor cross-sectional axis. The impact imparts an initial velocity to the system, which then oscillates.

Do the following.

- (1) Display input data.
- (2) Calculate the minor second moment of area I of the beam.
- (3) Calculate the minor cross-sectional stiffness EI .
- (4) Calculate the flex-beam spring constant k .
- (5) Calculate the initial velocity of the system $\dot{u}_0$ as resulting from the interaction between the projectile and the large mass.
- (6) Calculate the rad/s natural frequency $\omega_{n,\text{FBP}}$ and period of oscillation τ_{FBP} of the flex-beam pendulum with projectile embedded into it.
- (7) Calculate the response $u(t_1)$ after $t_1 = 10 \text{ s}$.
- (8) Plot the response of the system for $N_{\text{osc}} = 10$ oscillations and find on the plot:
 - (a) The largest response.
 - (b) The response value calculated at item (7).
- (9) Compare your results with those of Problem A.3.

C.9 Complex Exponentials and Trigonometric Functions

- (1) Find the solutions of $z^4 = 1$ (the fourth roots of unity) and express them as:
 - (a) Complex numbers.
 - (b) Complex exponentials.
- (2) Write the complex number $e^{i\omega t}$ as a sum of trigonometric functions (hint, Euler's formula).
- (3) Calculate the phase difference ϕ in radians and degrees between the following signals:
 - (a) Compare $x_1(t) = \sin(\omega t)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.
 - (b) Compare $x_1(t) = \sin(\omega t + \pi/3)$ and $x_2(t) = \sin(\omega t + 3\pi/4)$.
 - (c) Compare $x_1(t) = \sin(\omega t + 45^\circ)$ and $x_2(t) = \sin(\omega t + 115^\circ)$.
- (4) What is the phase in radians and degrees of the following complex numbers:
 - (a) $e^{i\pi/2}$.
 - (b) $e^{i3\pi/2}$.
 - (c) $e^{i\pi/4}$.
 - (d) $e^{-i\pi/2}$.
 - (e) $e^{-i\pi/4}$.
- (5) Draw the locations on the unit circle of the complex numbers listed in item (4).
- (6) Calculate the magnitude and phase in degrees of the complex number $Z = 4 + 5i$.